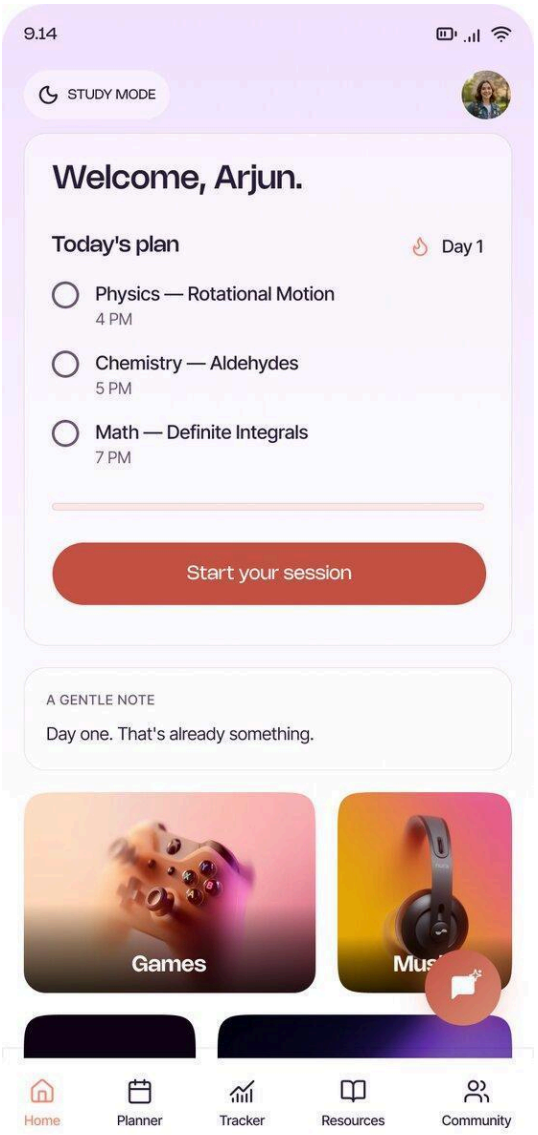
Headroom

A study companion with an embedded emotional support layer for Indian competitive exam aspirants aged 15–19.

Master Case Study — T4_05 · Signal Syndicate · Team 04 · Hackathon 8.0 · June 2026



SECTION 01

Project Overview

Headroom is a study companion app with an embedded emotional support layer, built for Indian competitive exam aspirants aged fifteen to nineteen.

On the surface, it looks like a planner. Today's tasks. Hours studied. Subjects on track. Inside that surface — hidden in plain sight — sits a quieter system: a peer-voice chat called Buddy, a write-and-release space called Vent, a contextual recovery flow called Burnout Reset, and an AI doubt-clearer called Ask AI. None of it is labelled mental health. None of it asks the student to identify as someone struggling.

This positioning emerged from a clear research finding. Students preparing for NEET, JEE, CA, UPSC, and similar exams carry intense, unpredictable pressure that peaks alone and at night. They reject anything labelled wellness, mindfulness, or mental health. But they will open a study app — and parents will approve of one. Productivity is the only stigma-free entry point.

Headroom is built on that insight. Every screen, every word, and every feature is designed to earn its place in a student's daily routine without naming the support layer underneath. The brand voice — Glass Sanctuary — is calm and direct, never therapeutic, never preachy. Buddy speaks in a same-age peer tone. Ask AI explains like an older sibling who's been through the syllabus. Burnout Reset offers a softer plan when the system detects missed sessions, instead of guilt-trip notifications.

The product was designed and prototyped during Hackathon 8.0 by Signal Syndicate, Team 04 — a distributed team of ten across India. Research, design system, prototype, and film were built end-to-end in eight weeks. This document walks through how we got from the problem to the product.

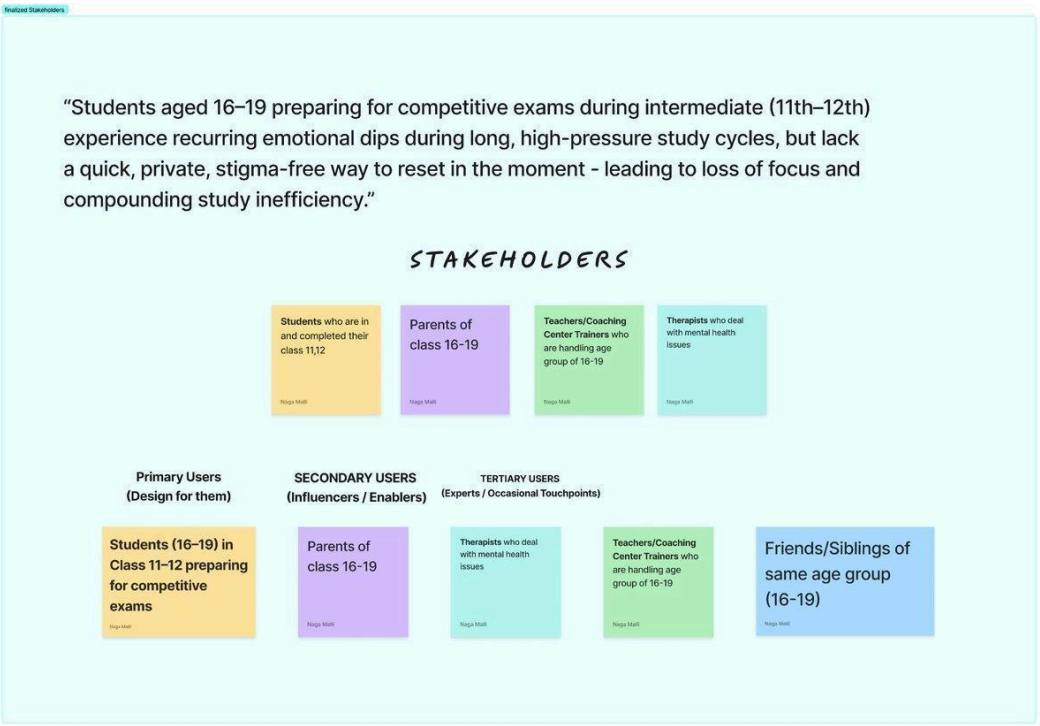
SECTION 02

Opportunity Discovery

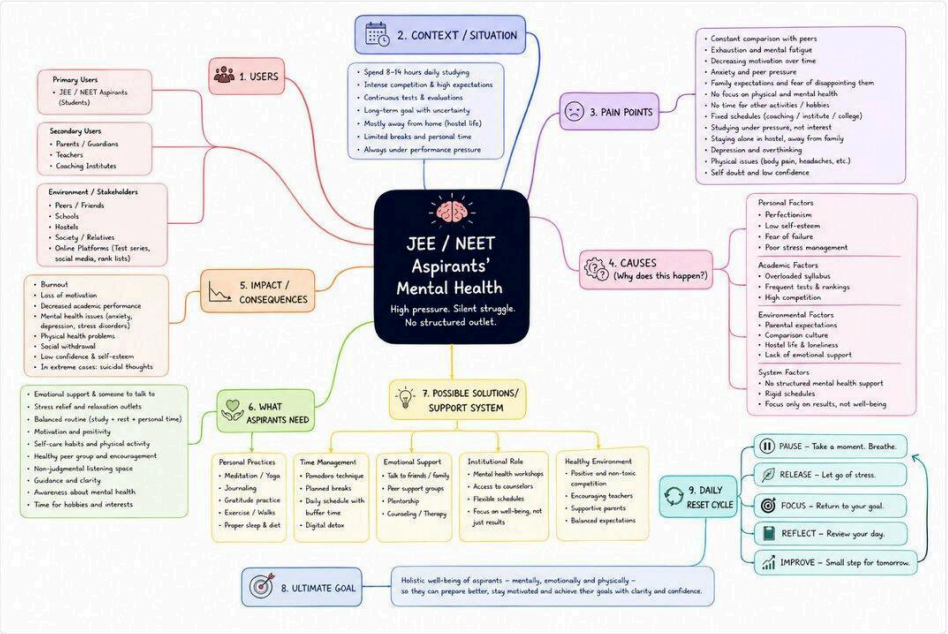
Before we knew what to build, we needed to know what we were looking at.

The opportunity discovery phase began with a broad question — what does emotional pressure actually look like for Indian competitive exam aspirants, and where do the existing solutions fall short. We mapped the ecosystem before we mapped the product. Primary users sit at the centre — students aged fifteen to nineteen preparing for NEET, JEE, CA Foundation, UPSC Foundation, or CUET, mostly in Class 11–12 or in dummy schools. Around them sit secondary stakeholders who shape the experience but don't use the app — parents, coaching faculty, peer groups, hostel environments, online test series, and rank lists. Beyond those, tertiary stakeholders — therapists, counsellors, and platforms that already serve adjacent needs.

Mapping the ecosystem surfaced four gaps that shaped the rest of the project.



Stakeholder & ecosystem map — primary, secondary, tertiary layers



Mind map — JEE/NEET aspirants mental health landscape

Four gaps that shaped everything

The stigma gap. Students openly reject anything labelled mental health, mindfulness, wellness, or self-care. Multiple students said they would not open such an app, would not let parents see it on their phone, and would not engage even if it was free. But the same students engaged readily with productivity apps, study planners, and AI-powered doubt-clearing tools. The category that's available has the right audience but the wrong label.

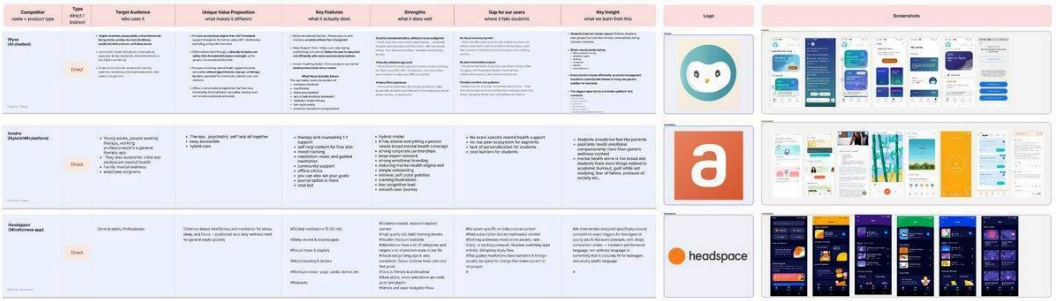
The pre-crisis support window. Existing support — helplines, therapy, school counsellors — is structured around crisis. A student has to identify themselves as struggling, reach out, schedule, and wait. The window where lightweight, in-the-moment intervention could help most — late-night anxiety after a bad mock test, repeated checking of rank groups, avoidance and backlog buildup — is exactly the window where nothing is available.

"No one reaches them" gap. When pressure peaks, students are typically alone. Helplines don't answer reliably at night. Therapy is unaffordable for most. Generic AI tools feel mechanical. Parents are often part of the pressure. Friends are in the same situation.

The productivity-app blind spot. The apps students do use — study planners, tracker tools, AI tutors — assume students are always mentally ready to study. They have no way to respond when a student is unable to start, falling behind, or avoiding. Wellness apps treat the academic context as out of scope. Nothing existing holds both.

> These four gaps together pointed toward a specific opportunity space — a product that operates inside the study routine, intervenes before a crisis, speaks in a language students don't reject, and is shaped for the exact context of Indian competitive exam preparation.

Competitor landscape



Competitor analysis — Wysa, Amaha, Headspace



Competitor analysis — Tele-MANAS, Woebot, Mitsu



Competitor analysis — Youper, Allen/Aakash, YouTube

SECTION 03

Research

Research was the longest phase of the project, and the one the entire product was built on. The goal was simple — understand the emotional reality of Indian competitive exam preparation directly from the people inside it, not from assumptions.

Starting wide, then narrowing

The initial problem statement was broader than what we eventually committed to. We started with an undefined age range, looking at the entire competitive-exam-preparing population in India. By the end of the first week of field research, the data was telling us something specific — students above 19 carried noticeably more evolved coping mechanisms, more self-awareness, and more developed support-seeking behaviours. The pre-crisis intervention window we were designing for closed earlier in their lives, not later. We narrowed the target population to ages 15 to 19 — the window where the pressure is sharpest and the coping infrastructure is least developed. Everything downstream was built against that narrowed brief.

Method

We ran 27 research touchpoints across three Indian states — Maharashtra, Telangana, and Karnataka. The mix was deliberate: 17 in-depth qualitative interviews and 10 structured surveys. We spoke with 20 students currently preparing for NEET, JEE, CA Foundation, UPSC, and SSC, sitting across coaching hostels, dummy schools, and home setups. We interviewed three parent households to understand the pressure environment around the student. We spoke with two educators — one coaching faculty member, one school teacher — to triangulate. And we spoke with one mental health professional, Samriddhi Deshmukh, whose knowledge of the field and firsthand experience with students personally known to her anchored the most important pattern in the data.

Interviews were conducted in person where possible and over video where not. Field photos from those sessions are included in this document — captured in homes, cafes, and online sessions, not in labs. The tone of the conversations mattered. We did not lead with mental-health framing. We asked about a typical study day, the WhatsApp groups they were part of, the moments they remembered most clearly from the last mock test, how they handled bad days, what they did when they couldn't sleep. The emotional material surfaced on its own, often within the first ten minutes.

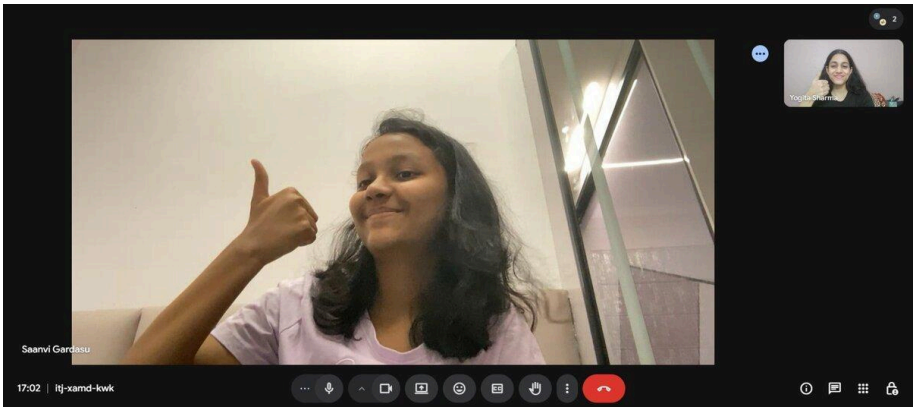
Field research — in homes, cafes & video calls



In-person interview session



Parent household interview



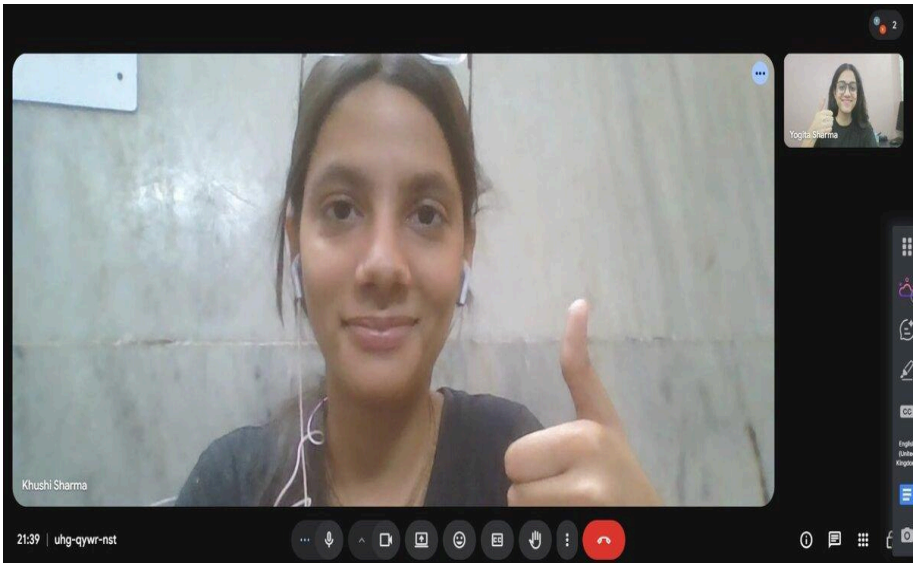
Home study session



Student interview at home



Video interview — Saanvi



Video interview — Khushi



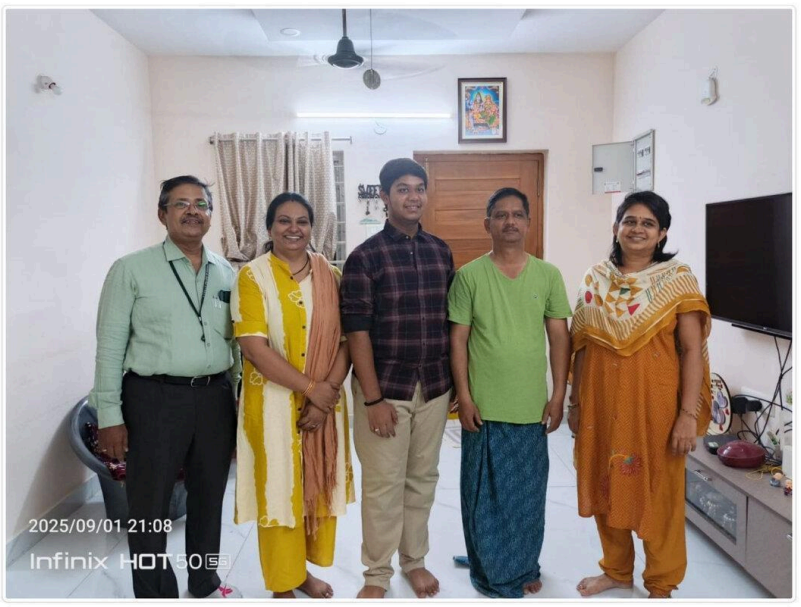
Home interview session



Parent interview



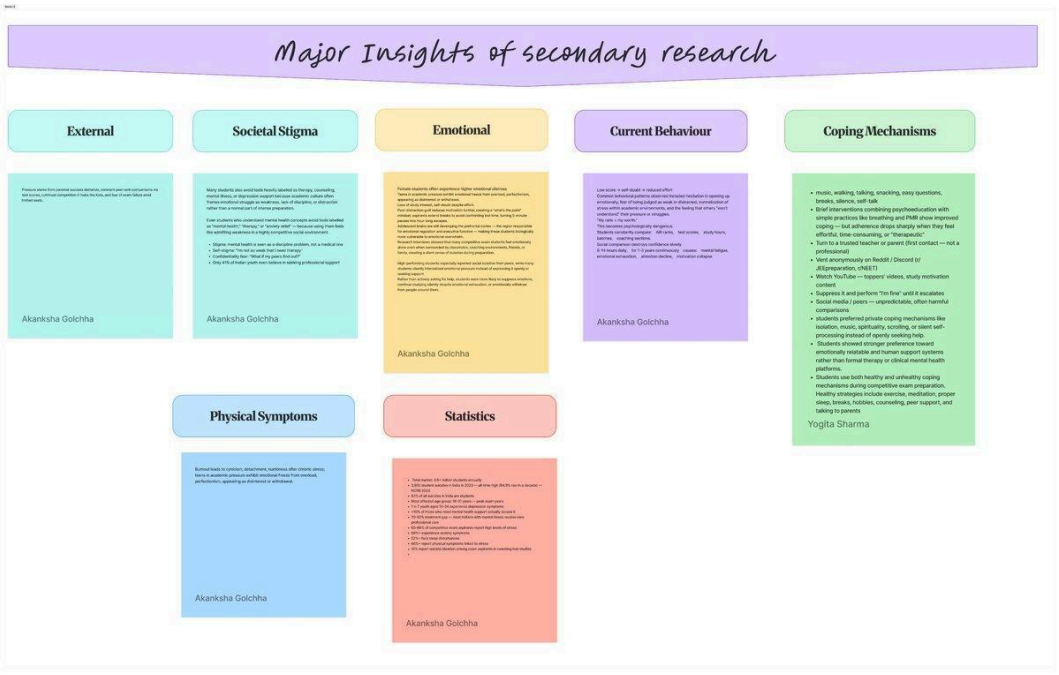
Field research — student interview



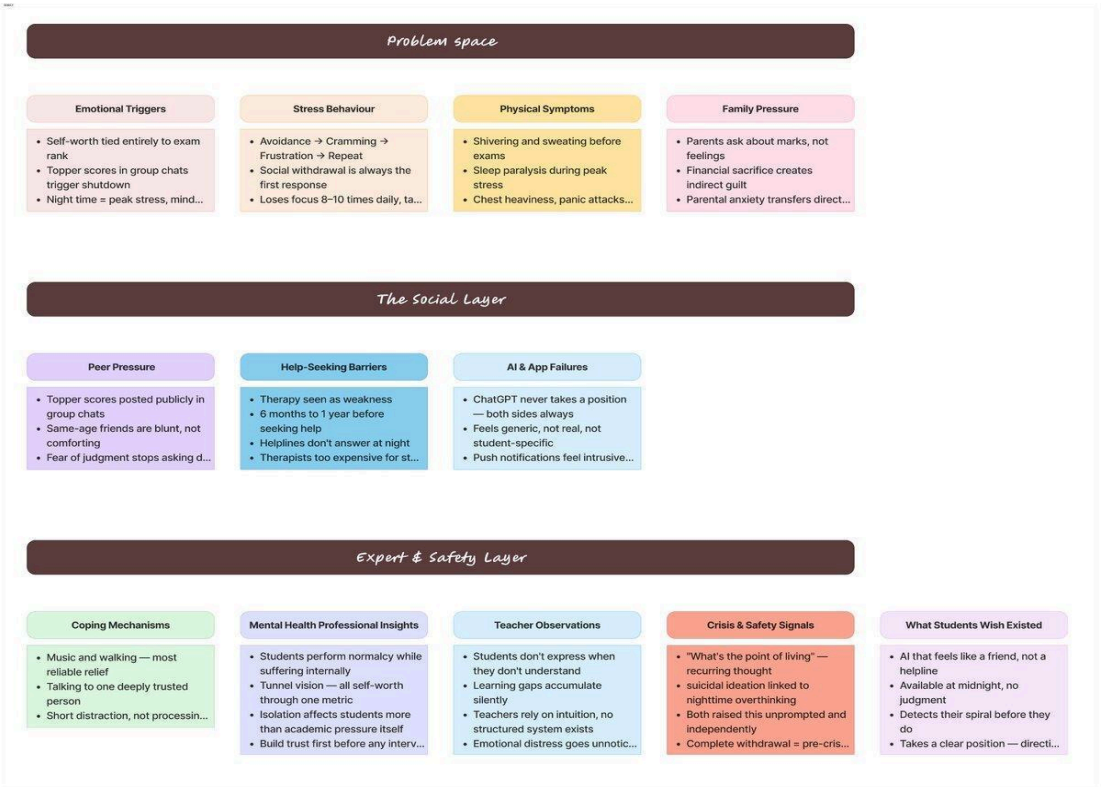
Parent household interview

What the secondary research said, and what the primary research confirmed

Secondary research established the scale and shape of the problem. Studies on Indian competitive exam aspirants documented elevated rates of anxiety, depression, sleep disruption, and suicidal ideation across the NEET/JEE preparation population. Government helplines like Tele-MANAS exist but are under-utilised by this age group. Wellness apps targeting Indian users — Mitsu, Wysa, Amaha — focus on adult populations or general anxiety, not the specific pressures of competitive exam culture.



Secondary research — major insights across 7 dimensions



Five patterns that shaped the product

Pressure peaks alone, at night. Across interviews, the hardest moments arrived between 11 PM and 2 AM — after the day's study was done, after the mock test scores had been shared, after parents had gone to sleep. Helplines didn't answer. No one was available. Several students described the same loop: scrolling test-centre WhatsApp groups, comparing scores, then closing the phone and not sleeping.

Comparison happens in real time. Mock test scores are now shared in WhatsApp groups within minutes of test completion. There is no recovery window between getting a result and seeing where you fall in the cohort. Multiple students said this single feature of their environment was the most consistent trigger for emotional dips.

Rigid plans punish missed days. A student who misses one or two study days does not return to the plan the next day. The backlog produces guilt, the guilt produces avoidance, the avoidance produces more backlog. Every student we spoke to had been through some version of this loop.

The support system is often the pressure. Parents, coaching faculty, and peers were named as sources of pressure as often as they were named as sources of support — sometimes both, by the same student in the same conversation.

Students reject anything labelled mental health. The single most consistent rejection across our entire sample was of products, services, or conversations framed around mental health, wellness, or self-care. Two students explicitly said they would uninstall any app on their phone if a parent saw it labelled that way.

> ⚠ One unexpected finding: across interviews, multiple students surfaced thoughts of self-harm unprompted — not as a current intent, but as a pattern of thinking that had visited them at least once. Samriddhi confirmed this is consistent with what's seen broadly in this age group. This finding directly shaped the decision to keep Emergency support outside the paywall, on the Home screen, accessible at any time.

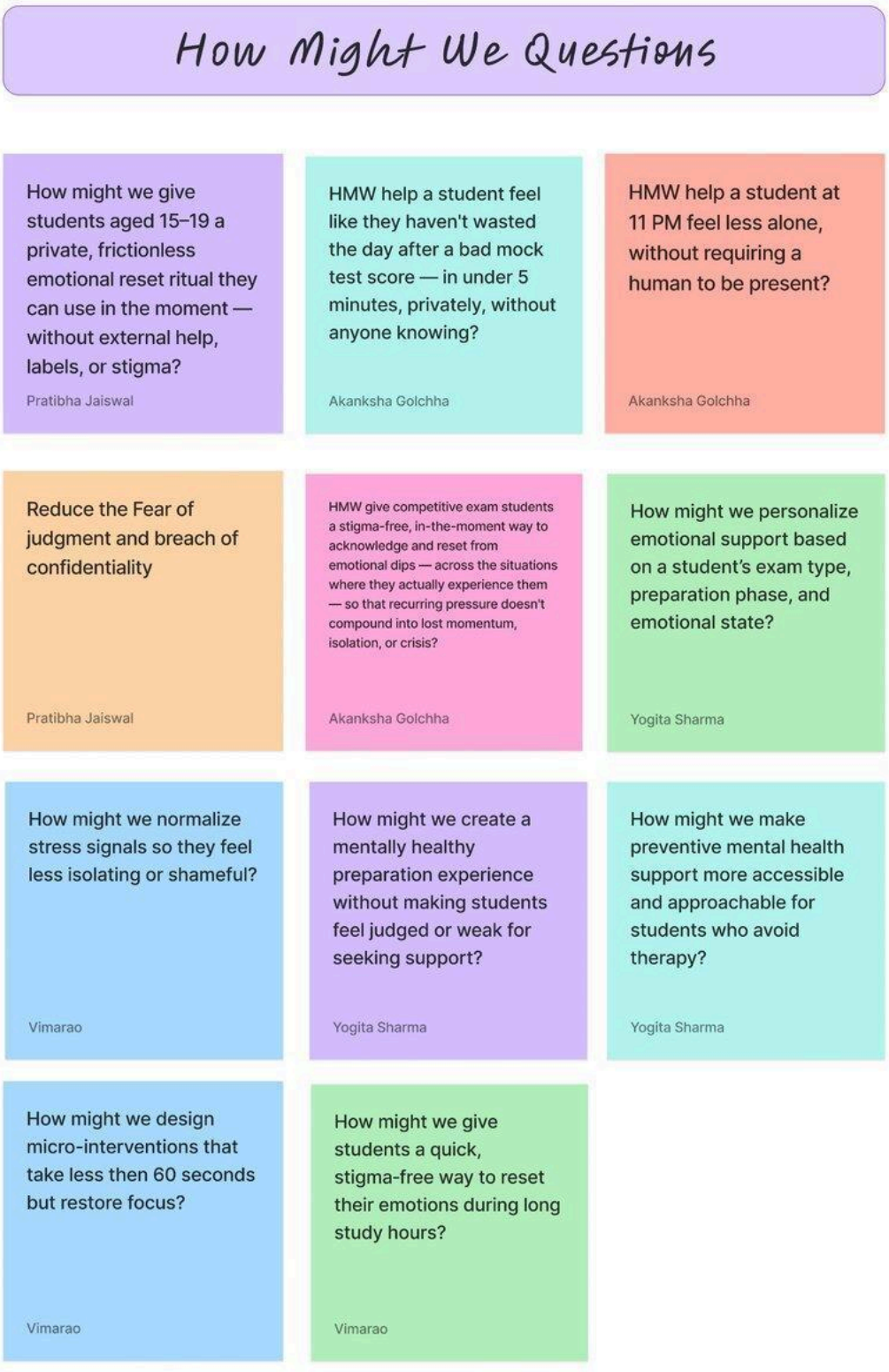
SECTION 04

Ideation

Research gave us the shape of the problem. Ideation was where we worked out the form of the answer.

Diverge

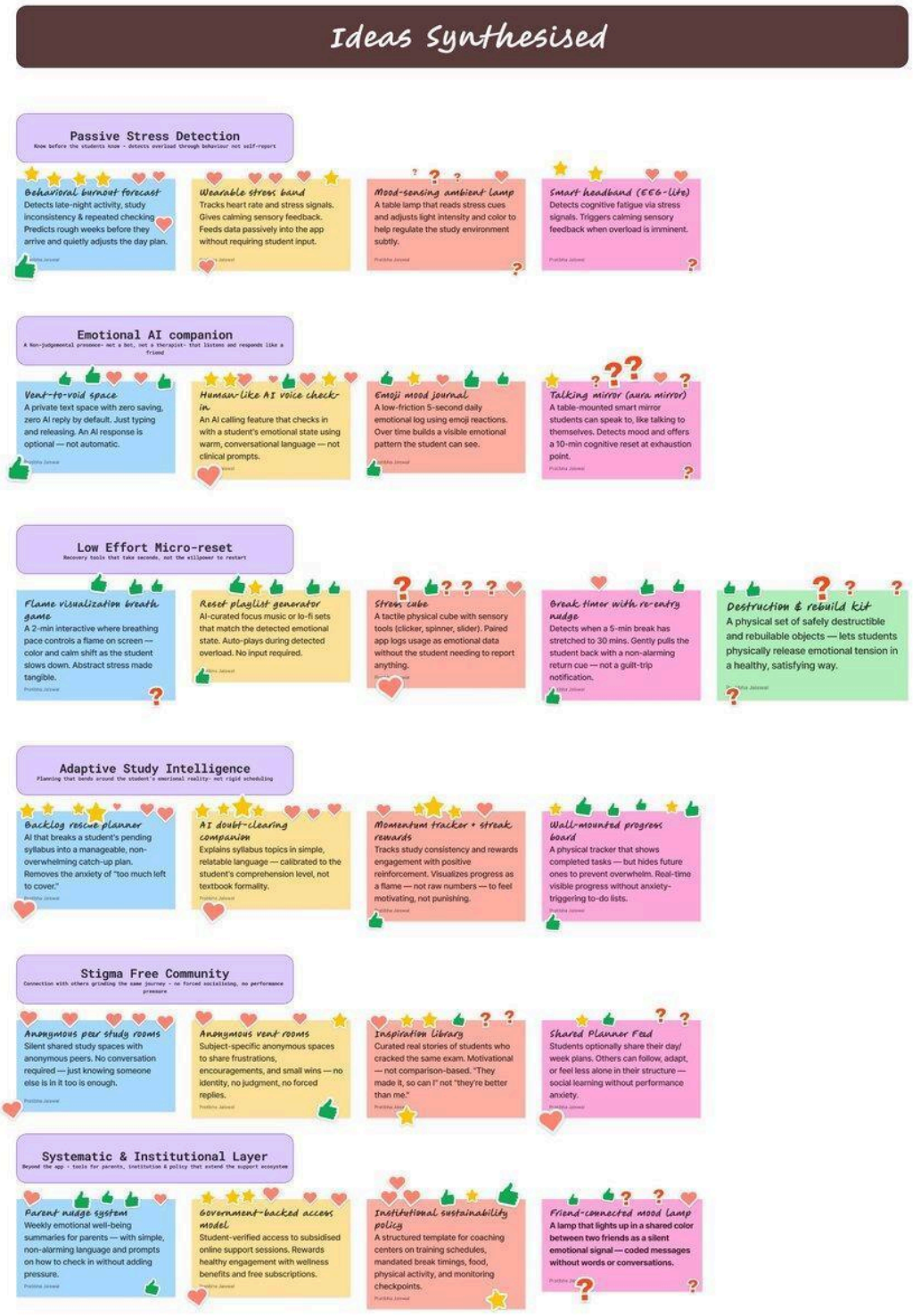
We started wide. The full team ran a divergent brainstorm using How-Might-We questions framed against each gap surfaced in research. From these prompts, twenty-eight distinct ideas surfaced across six thematic territories: Passive Stress Detection, Emotional AI Companion, Low-Effort Micro-Reset, Adaptive Study Intelligence, Stigma-Free Community, and a Systemic Layer covering parents, institutions, and policy. We deliberately didn't filter early.



HMW board — 11 questions across the problem space

Convergence — seven steps

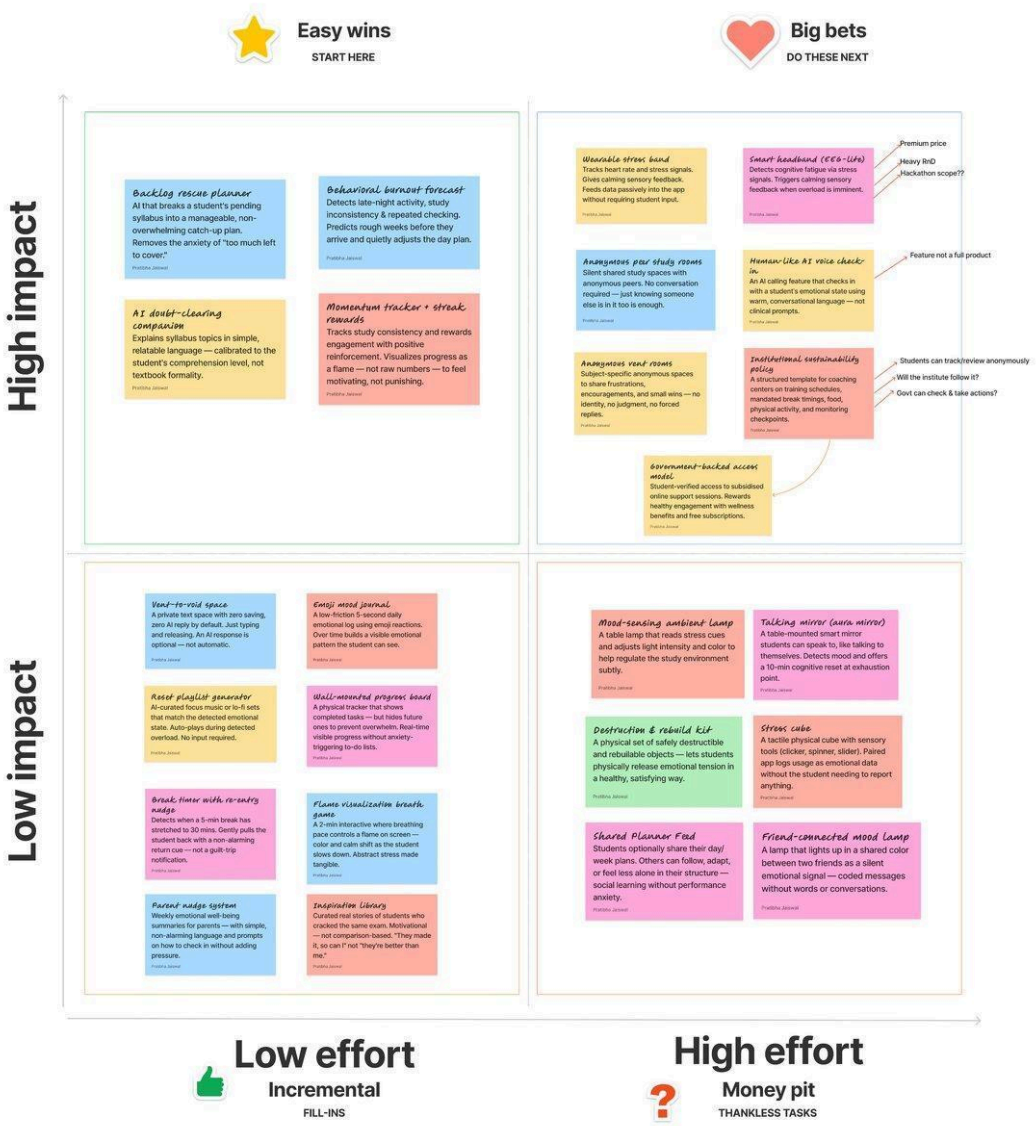
Step 1: Converge. Broad consolidation of the divergent ideas. Duplicates merged, edge cases noted, the wildest concepts kept on the board with question marks rather than discarded outright.



Ideas Synthesised — six clusters

Step 2: Synthesise. The converged ideas were merged into common categories — six clusters that grouped ideas by the kind of problem they solved, not the kind of feature they were.

Step 3: Impact-vs-Effort matrix. Each idea was placed against two axes — how much it could move the needle for our user, and how much it would cost in time, technology, or risk to build.



Impact vs Effort matrix — Easy Wins vs Big Bets

Step 4: Top 3 Concepts. Three rose to the top: Study Planner/Companion/Tracker, a Government-Backed Policy Tool, and a Wearable Headband.



Top 3 Concepts voted by the team

Step 5: Select 1. The Study Planner/Companion/Tracker concept was selected.The Government-Backed Policy Tool was ruled out for hackathon scope. The Wearable Headband was ruled out for hardware R&D needs.

Step 6: Potential Features. Against the selected concept, we mapped out the full possible feature set.

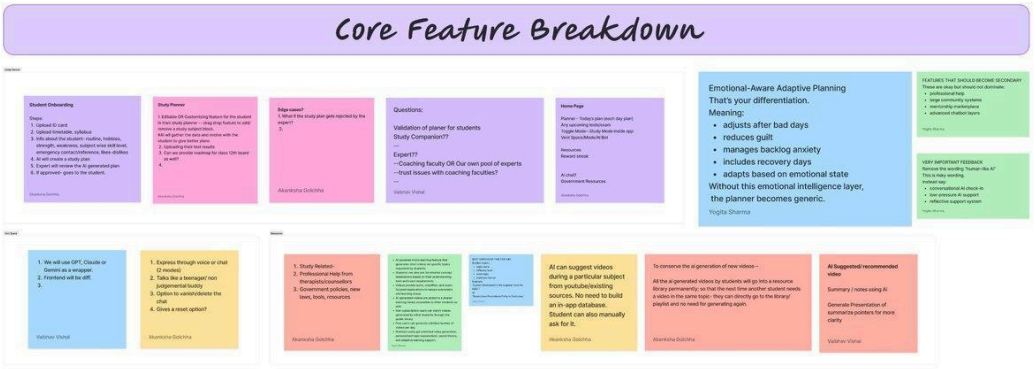


Potential features mapped against selected concept

Step 7: MoSCoW. Must-Haves anchored the product: study planner, vent space, AI buddy chat, AI doubt-clearing, momentum tracker, backlog rescue, habit builder, community, professional help. Should-Haves: government-backed access model and break-timer re-entry nudge. Could-Haves: AI-curated reset playlists and an ebook/resource library.



MoSCoW board — Must / Should / Could / Won't Haves



Core feature breakdown

The Trojan Horse moment

> The product on paper still looked like a wellness app with a study layer bolted on. The reframe came in a team discussion — what if we inverted the framing entirely? What if the product looked like a study app on the surface, and the emotional support layer lived hidden inside, never named as mental health? That single move — the Trojan Horse positioning — changed every downstream decision.

Naming

From a long list of explorations — OnPace, Headroom, Methodroom, Stride, Rebound, Focus Flow, and several others — the team converged on Headroom. The name carries the meaning we needed: the space between the student and the pressure pushing down on them. Not the absence of pressure. The space to breathe inside it.

Name	Meaning / story	Vibe	Brand promise it carries	Best line to pitch it
OnPace	"On pace" — a runner's term for being on track with their goal. The app helps you stay on your own pace, not someone else's.	Calm-competit ive, clean, athletic	Steady forward motion at your own rhythm	"You don't have to be the fastest. Just stay on pace."
Pacecore	Pace + "-core" suffix. "Core" is a Gen Z aesthetic marker (studycore, cottagecore). Your pace is your core — your steady centre.	Gen Z aesthetic, modern, trend-aware	Pace as identity — this is who you are, not just what you do	"Find your pacecore. Build your study era."
Strideclub	Strideclub — a club for people taking it in stride. Quietly social. The idiom "to take something in stride" means handling pressure with grace.	Warm, community-for ward, peer-energy	You're not alone in this — others are pacing alongside you	"Stride your stride. Together."
Inphase	Physics term: two waves perfectly synchronised are "in phase." To be in phase is to be in sync with yourself — your energy, your study, your rhythm.	Distinctive, smart, quietly poetic	Sync with yourself, not the noise around you	"Be in phase with you."
Methodroom	A room where you practise your method. Method = your way of doing things; Room = a private space where it's safe to be yourself.	Quiet, private, editorial	A safe, personal space to build your own way of studying	"Your method. Your room."

Uncrumple
Refound
Headroom
Rootspace
iBreathe

Yogita Sharma

1. Ascend
Meaning: Moving upward despite challenges.

2. Udaan
Meaning: Flight, rising higher.

3. ReFocus
Meaning: Getting back on track after distraction.

4. Sankalp
Meaning: Dete mination or commitment.

tanuj gahtori

Rebound
Focus flow
Zenprep
(Exam prep with inner calm)
iBreathe

Nook -- a small sheltered corner — a quiet pocket of space. Often warm, partially enclosed, and meant for one person — somewhere you go to read, think, or just be.

Ease — Modern, soft
"Find your ease." "Studies with ease."

Lume — From luminous. Soft glow

OnStride-- Carries the resilience meaning natively (the idiom "to take something in stride")

Akanksha Golchha

Nova
Meaning: A star that shines brightly again. Symbolizes recovery after burnout.

Evora
Meaning: Invented brand name (evoke + aura). Sounds premium and app-worthy.

Milo
Friendly and approachable.

Dhairya
Meaning: Patience & resilence

Shunya
Zero or stillness. It matches perfectly with the cognitive load of micro-reset games.

Pratibha Jaiswal

PREP PLUS (THE HEART BEAT OF EXAM)
REBOUND (RECOVER AFTER A SET BACK)
MENTRA (MENTAL + MANT
CALMERA
VIORA
FOCUS FLOW
EUNOIA (GREEK WORD FOR PEACE)

Anandu Sv

Headroom

A student is constantly surrounded by exams, expectations, competition, comparisons, and stress. When that pressure becomes too much, they lose clarity and feel overwhelmed.
Headroom means creating enough space in their mind to breathe, think clearly, stay balanced, and perform at their best.
In one line:
Headroom is the space between a student and the pressure pushing down on them.

Yogita Sharma

OnPace

A student is constantly told to go faster — more hours, more revisions, more sacrifice. Everyone around them is racing: classmates, coaching toppers, family expectations. In that noise, the student loses their own rhythm and either burns out trying to match someone else's pace, or falls behind and shame-spirals. OnPace means moving at the rhythm that actually works for you — not too fast, not too slow, just steady. The pace that's sustainable, the pace that's yours. Sustainable. Matched to who you are.

Akanksha Golchha

App naming exploration board

SECTION 05

UX Architecture

With the concept locked and features prioritised, the next phase was translating intent into structure. UX Architecture covered four parallel tracks — defining who we were designing for at a person level, mapping the emotional journey they would take through the product, structuring the information system, and laying out the screens.

The persona

We built a primary persona, Sarvani Sharma — 17 years old, Mumbai, NEET aspirant, Class 12, 6–8 study hours daily. The persona consolidated patterns from research: a disciplined student who measures self-worth through performance, suppresses emotions internally, pushes through burnout, prefers private and non-judgmental spaces, and rejects pep-talk-style motivation. Every design decision downstream was checked against her.



Name: Sarvani Sharma
Age: 17 Years
Location: Mumbai
Exam Preparation: NEET / Competitive Exams
Academic Stage: Class 12 Student
Study Duration: 6–8 Hours Daily

"People who haven't prepared for competitive exams don't really understand this pressure."

Sarvani is a disciplined and hardworking student preparing for competitive exams. She follows a strict study routine and silently handles emotional pressure, fear of falling behind, and constant academic comparison.

GOALS
- Crack a top competitive exam
- Maintain daily consistency
- Improve focus and productivity
- Make parents proud
- Stay ahead academically
- Balance preparation and mental stability

PREFERENCES
- Emotionally relatable support
- Private and non-judgmental spaces
- Short breaks without productivity guilt
- Realistic motivation instead of generic advice
- Structured and organized environments

KNOWN HABITS
- Studies even during exhaustion
- Follows strict routines
- Tracks performance constantly
- Revises lighter topics during burnout
- Uses short controlled social media breaks
- Avoids openly discussing stress
- Continues studying despite low focus

PAIN POINTS
- Fear of falling behind peers
- Constant topper comparison
- Guilt while resting
- Emotional exhaustion from continuous studying
- Anxiety during low-performance periods
- Difficulty recovering after setbacks
- Pressure from parents, teachers, and coaching culture

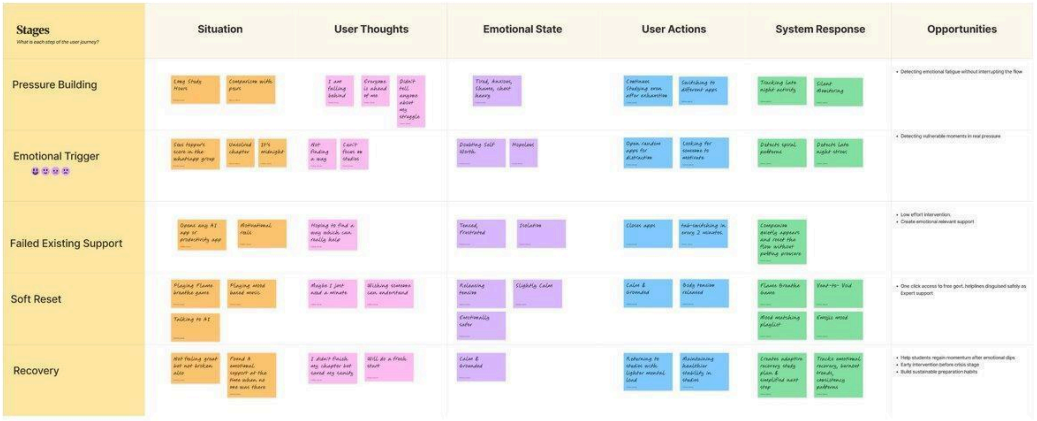
NEEDS
- Quick emotional reset support
- Consistency recovery during low phases
- Emotionally safe guidance
- Support that understands competitive exam pressure
- Small mental recovery tools integrated into study flow

BEHAVIOR
- Measures self-worth through performance
- Pushes herself despite burnout
- Suppresses emotions internally
- Relies heavily on discipline and routine
- Prioritizes productivity over emotional recovery
- Becomes emotionally withdrawn during stress

Primary persona — Sarvani Sharma, 17, NEET aspirant, Mumbai

The user journey

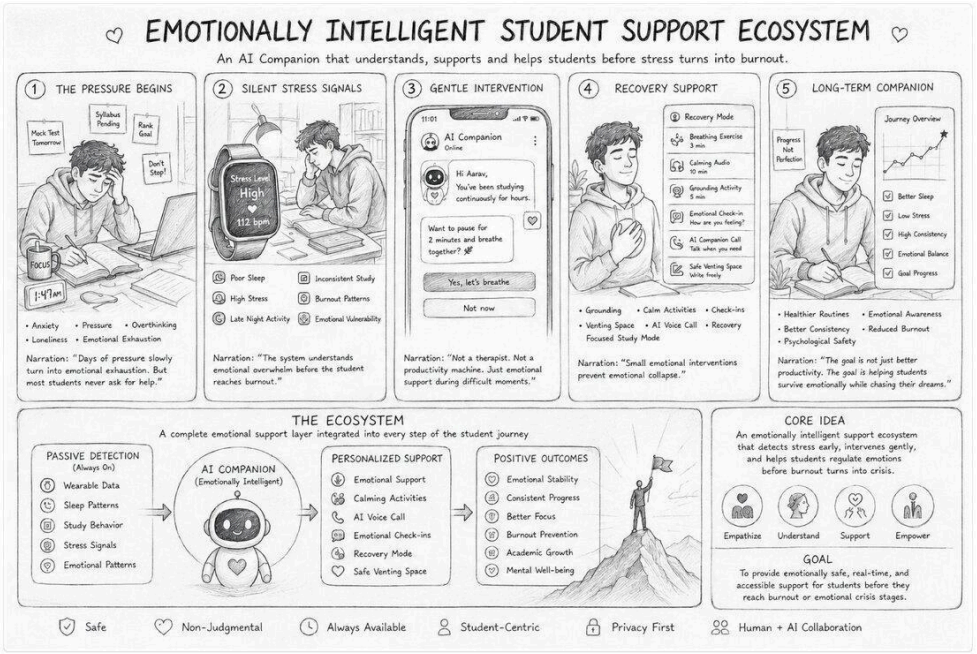
We mapped five emotional stages a student moves through during a typical bad cycle — Pressure Building, Emotional Trigger, Failed Existing Support, Soft Reset, and Recovery. For each stage we documented the situation, user thoughts, emotional state, observable actions, system response, and design opportunities. The journey map became the spine of the product.



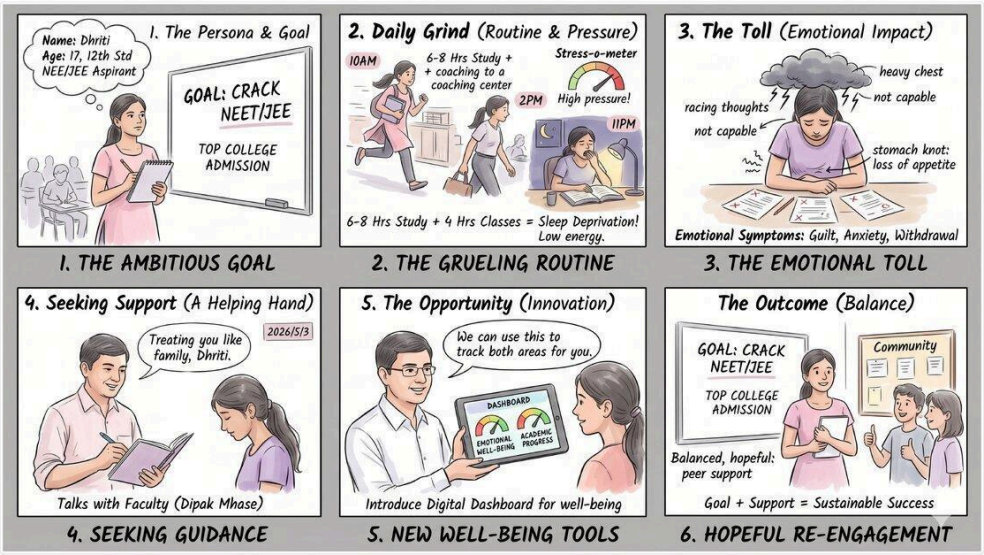
User journey map — five emotional stages across a pressure cycle

Storyboards

We produced three storyboards — one for the Smart Mirror concept that didn't survive, one for the AI Companion ecosystem, and one for a faculty-mediated dashboard concept. These were used early to communicate intent across the team and to test whether the emotional logic of our concepts held up in a narrative format before we wrote a single line of UI.



Storyboard — AI Companion ecosystem

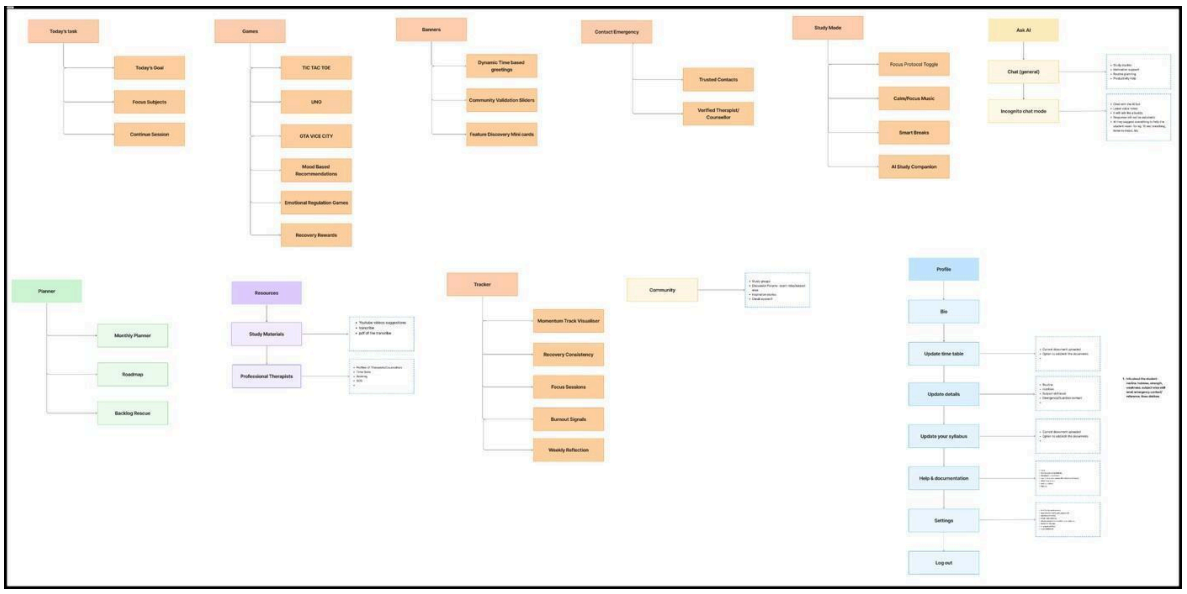


Storyboard — Faculty-mediated dashboard concept

Information Architecture

The IA settled around a five-tab bottom navigation — Home, Planner, Tracker, Resources, Community — with a persistent Ask AI access point and an Emergency entry never more than two taps from any screen. The architecture deliberately keeps the visible layer (Home, Planner, Tracker) productivity-shaped, while the embedded layer (Buddy via Ask AI, Vent via Home, Burnout Reset via contextual nudge) is hidden inside everyday surfaces.

Information architecture — five-tab navigation structure

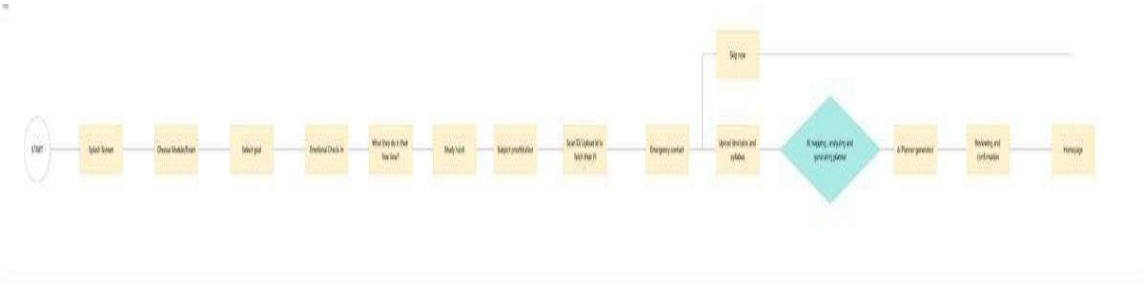


Information architecture — five-tab nav with embedded emotional layer

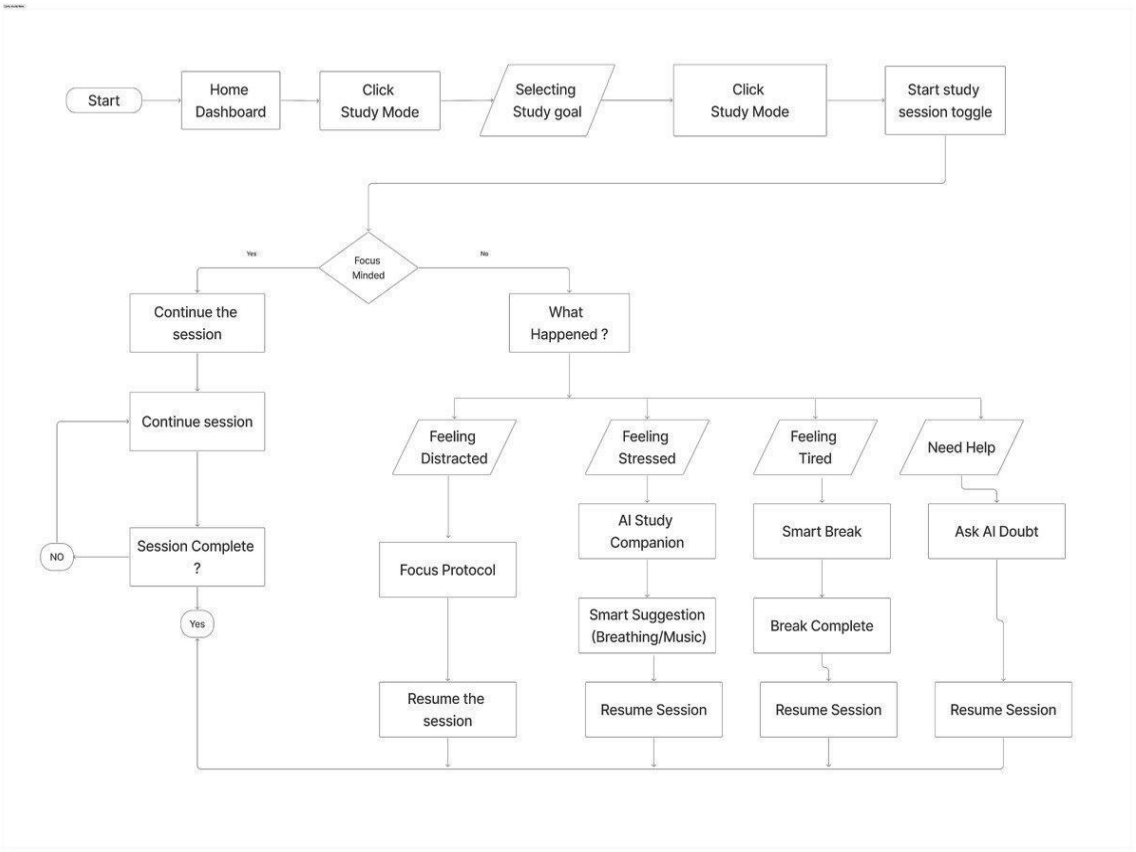
User flows

Seven core flows were diagrammed — Onboarding, Daily Study, AI Suggested Resources, General AI Chat, Incognito (Buddy) Chat, Burnout Detection, and User Profile. Each flow was reviewed for friction, drop-off risk, and emotional safety. The Incognito Chat flow received the most iteration — early versions

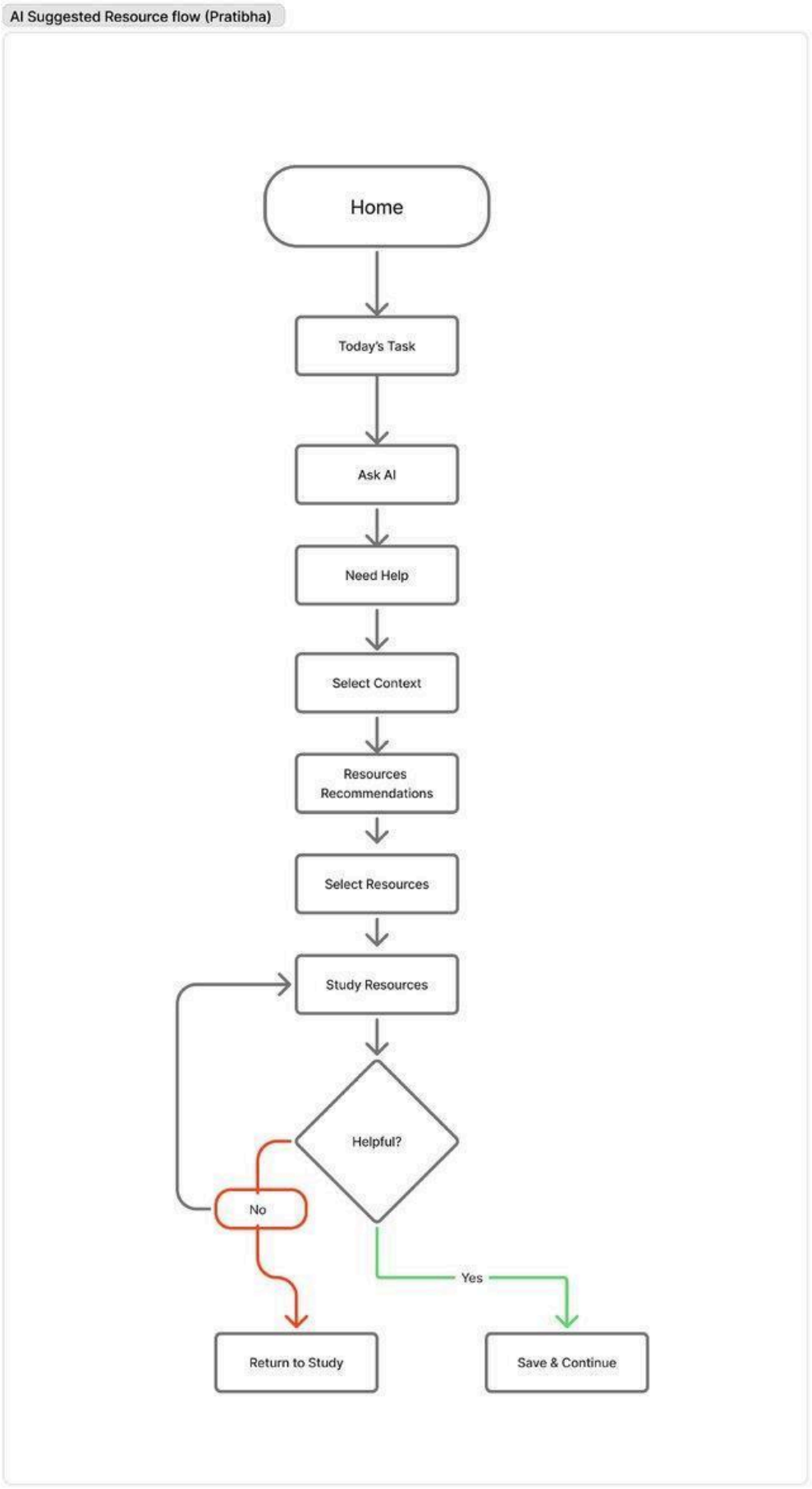
saved conversation history by default, which contradicted the safety principle. The final flow saves nothing, displays the privacy state visibly.



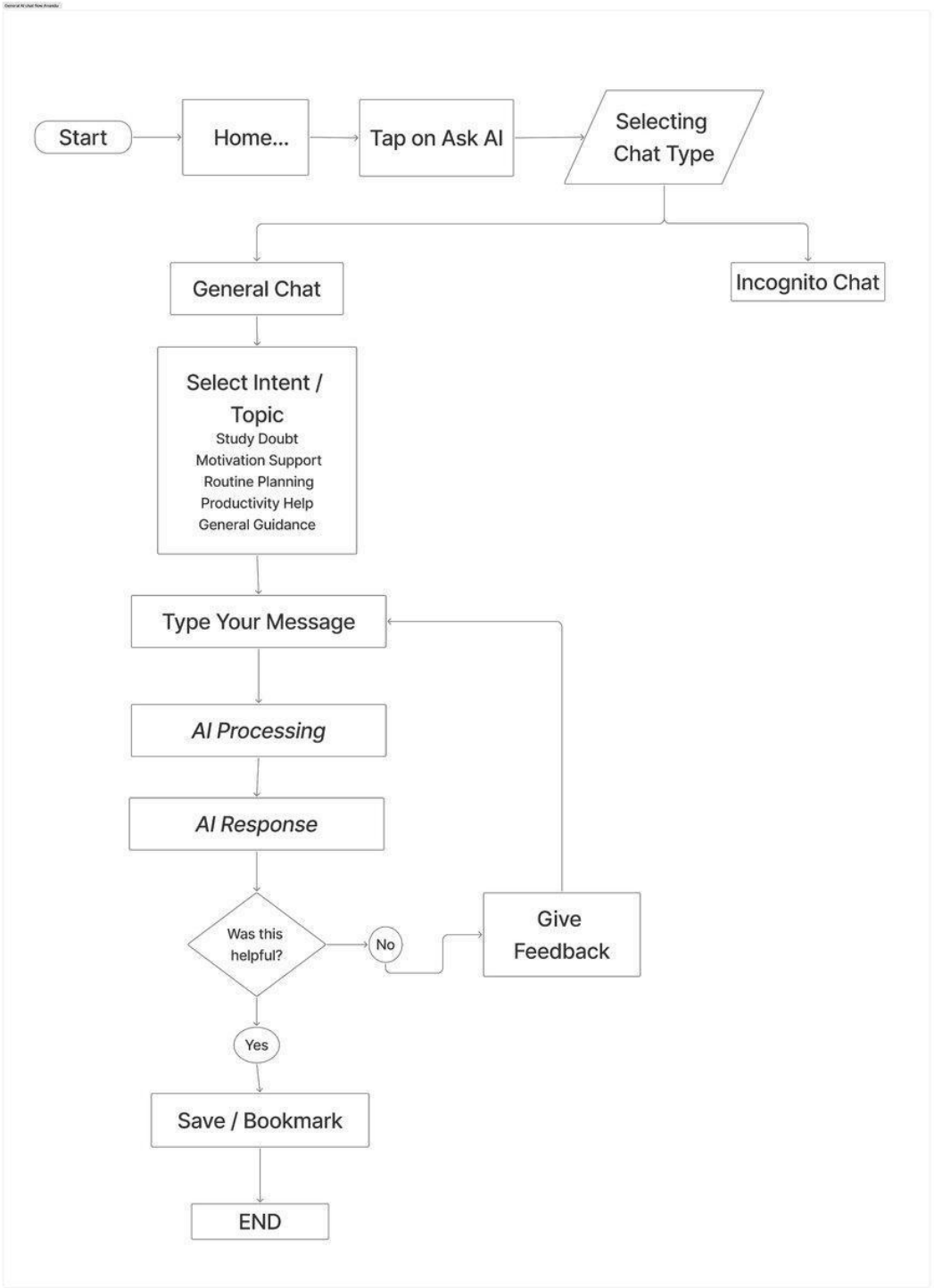
Onboarding flow — 12 steps from splash to home



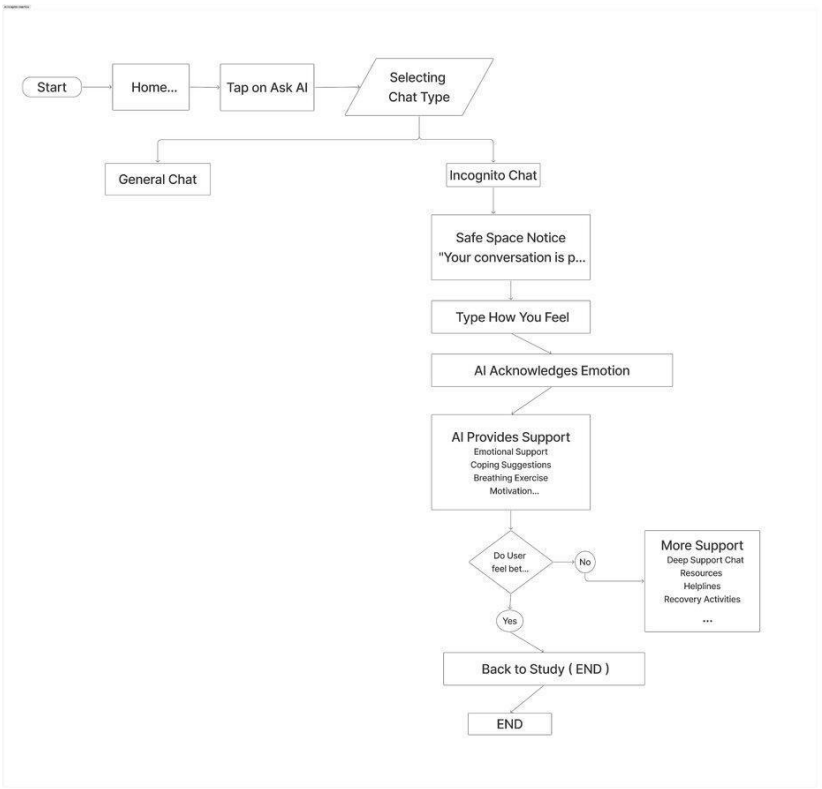
Daily study flow — session states and emotional branching



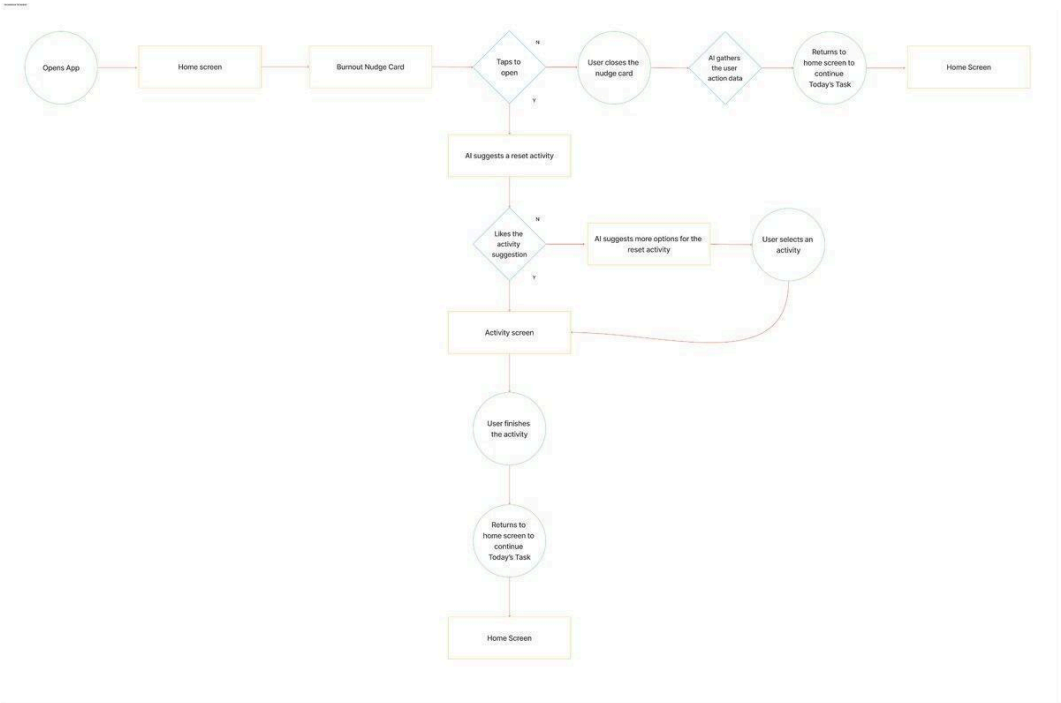
AI suggested resources flow



General AI chat flow



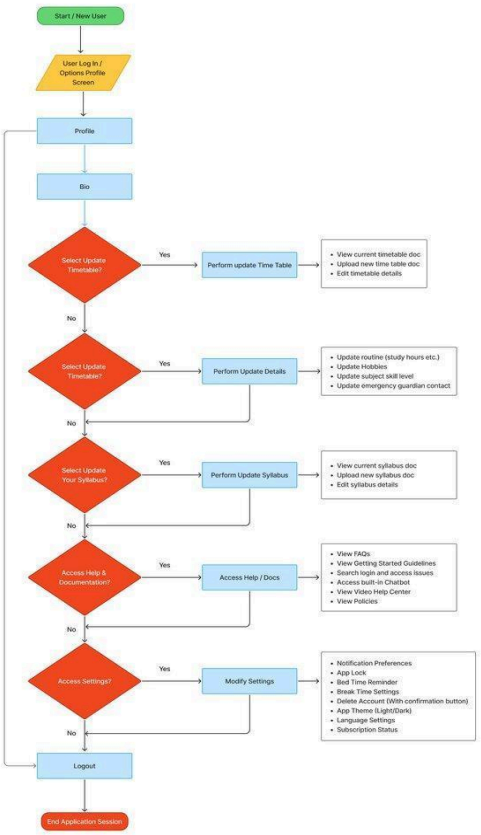
Incognito (Buddy) chat flow



Burnout detection flow

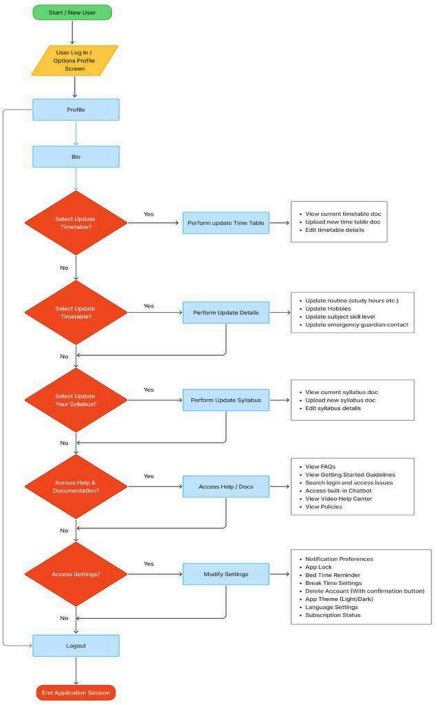
User Profile

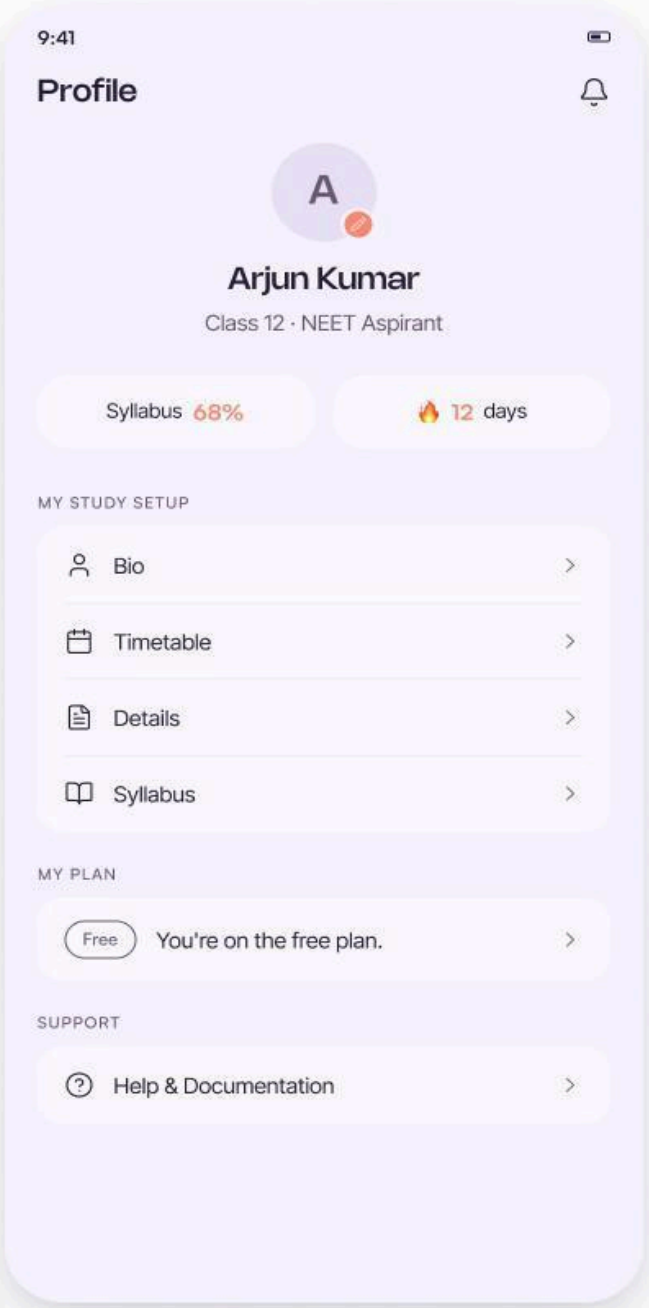
Profile → Bio → Update Time Table → Update Details →Update Syllabus → Help & Documentation → Settings → Log Out



User Profile

Profile → Bio → Update Time Table → Update Details →Update Syllabus → Help & Documentation → Settings → Log Out





Profile — Bio, Timetable, Syllabus, Plan status. All study context in one place.

User profile flow

Low-fidelity wireframes

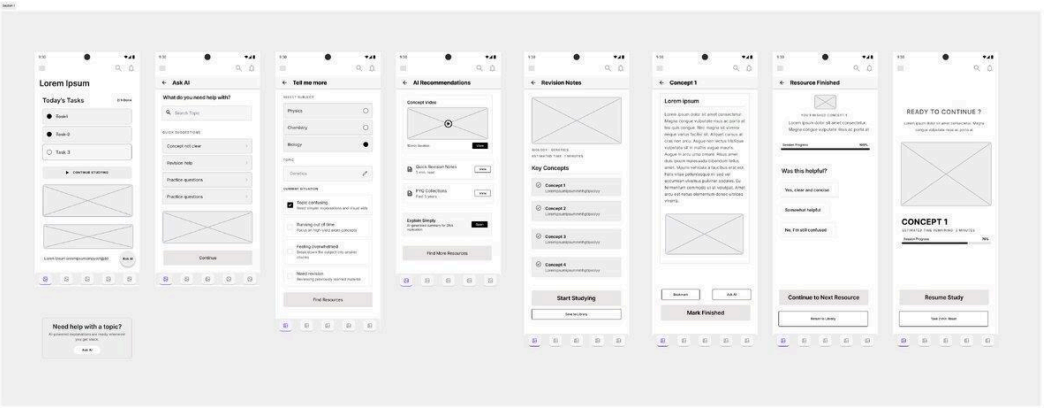
Five flows were taken to lo-fi: Daily Study, Onboarding, Profile, Ask AI, and Buddy Chat. Lo-fi was deliberately kept rough — boxes and labels only — so the team could critique structure without getting distracted by visual decisions that hadn't been made yet. Visual design started only after the structure was approved.



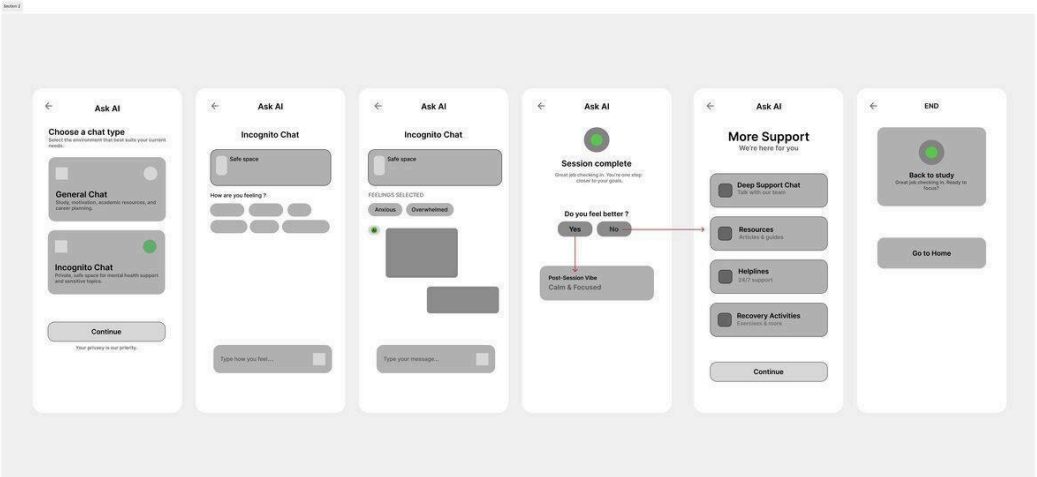
Lo-fi — Onboarding screens



Lo-fi — Daily Study flow



Lo-fi — Ask AI



Lo-fi — Buddy Chat



Lo-fi — Profile

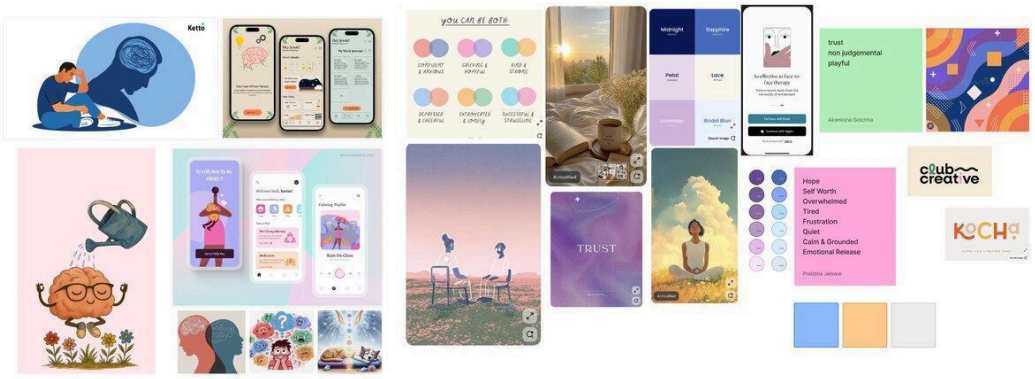
SECTION 06

Visual Design

Visual design was the layer where positioning became palpable. Every decision was checked against one question — does this look like a study app a parent would approve of, while feeling emotionally safe enough that a student would actually trust it?

Mood and direction

We started with a moodboard pulling references from mental wellness illustration, soft photography, calm digital interfaces, and the colour-emotion mapping we had built from research. The convergence point that emerged: warm, quiet, emotionally legible — never clinical, never childish, never aggressive.

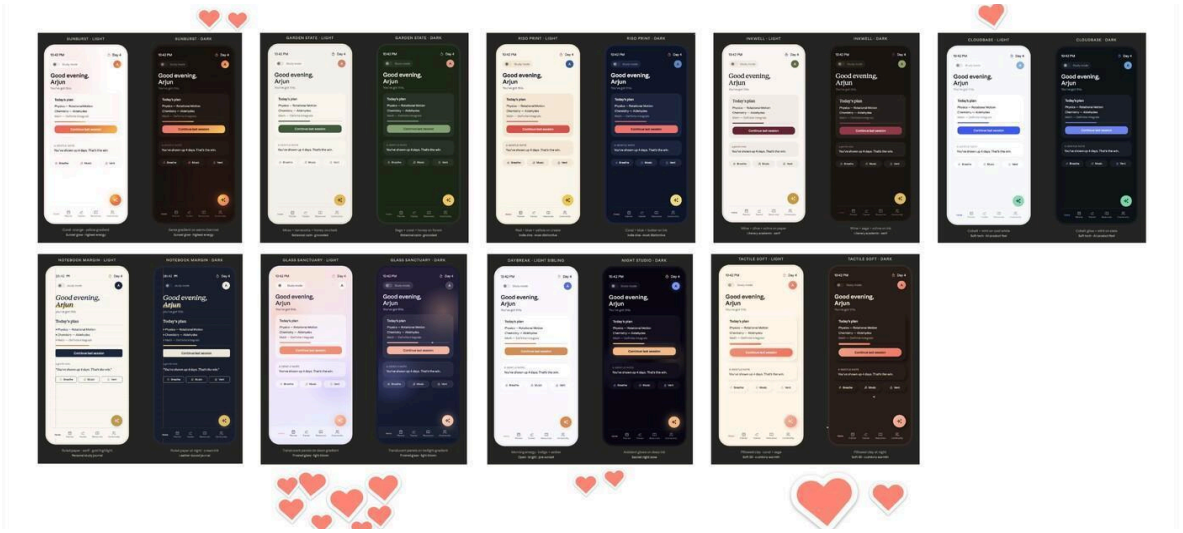


Visual moodboard — references and rejected directions

Brand system — Glass Sanctuary

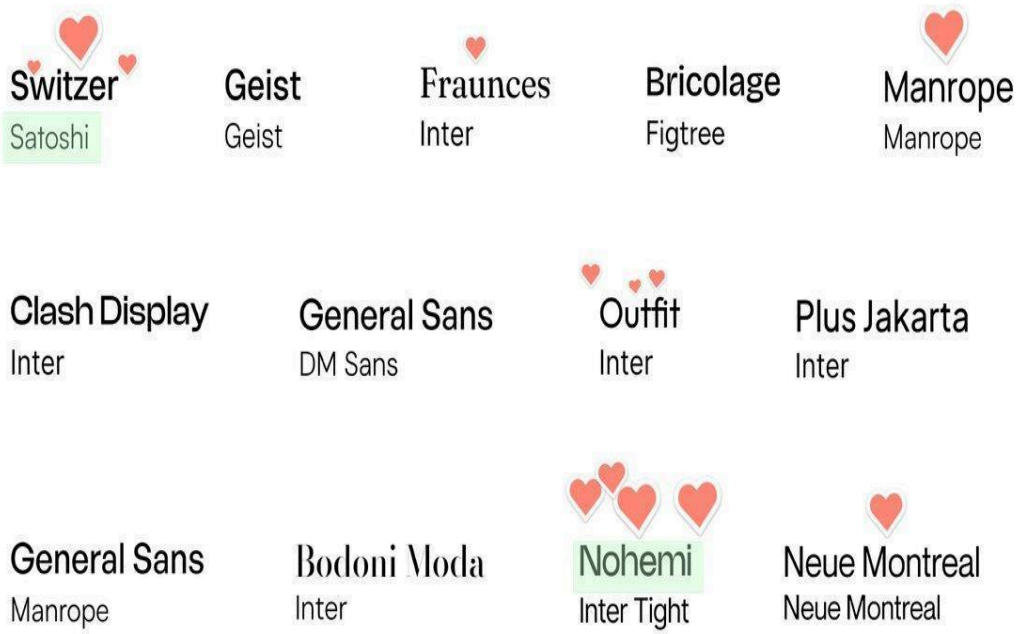
The visual language we landed on is called Glass Sanctuary. It rests on four ideas: frosted-glass surfaces that feel translucent and protected, a coral-and-aubergine palette that signals warmth without sentimentality, ambient gradient blooms in peach-to-lavender-to-sky on hero surfaces, and generous breathing room across every screen. The aesthetic is calm rather than soothing, considered rather than gentle.

App visual design direction — Glass Sanctuary theme variants



Typography

Type pairing came after an exploration of fifteen options across display and body candidates. We landed on Nohemi for display and Satoshi for body. Nohemi carries a quiet editorial confidence — readable at small sizes, distinctive at large sizes, never decorative. Satoshi handles body text with high readability across small screens and dense data layouts.



Font exploration — 12 pairings; Nohemi + Satoshi selected



Typography

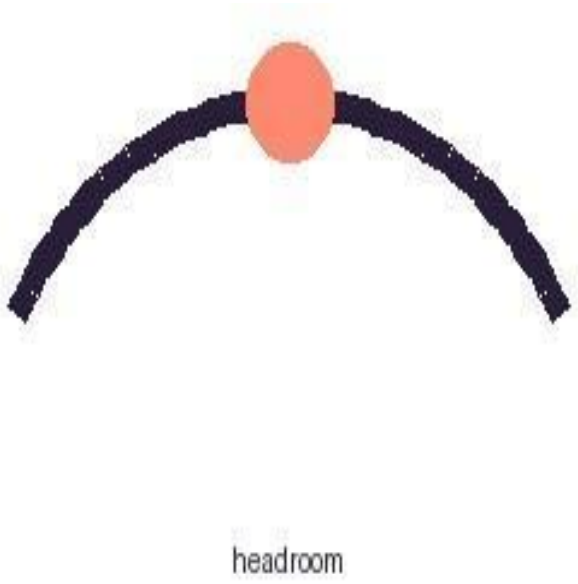
XL: Title/Header 1	Family: Nohemi Weight: 500 Size: 32px Line Height: 36px Letter Spacing: -0.5% Use: Splash screen,	Title/Header 1
LG: Header 2	Family: Nohemi Weight: 500 Size: 24px Line Height: 28px Letter Spacing: -0.3% Use: Home greeting, Never	Header 2
MD: Header 3	Family: Nohemi Weight: 500 Size: 20px Line Height: 24px Letter Spacing: -0.2% Use: Top Nav title on Home screen only (Avatar + Title + Action variant).	Header 3
MD: Header 3	Family: Nohemi Weight: 500 Size: 16px Line Height: 20px Letter Spacing: -0.1% Use: Top Nav title on all back-navigation	Header 3
SM: Small section headings	Family: Nohemi Weight: 500 Size: 14px Line Height: 18px Letter Spacing: 0% Use: Small section headings inside cards, modal titles.	Small section Heading
Body (lg)	Family: Satoshi Weight: 400 Size: 16px Line Height: 24 Use: Long-form text, Vent textarea content, AI response body text	Body
Body (md)	Family: Satoshi Weight: 400 Size: 14px Line Height: 20	Body
Body (sm)	Family: Satoshi Weight: 400 Size: 12px Line Height: 16	Body
Label-md	Family: Satoshi Weight: 500 Size: 12px Line Height: 16 Letter-spacing: 0.8 UPPERCASE Use: Card section labels — "TODAY'S PLAN", "A GENTLE NOTE", "PATTERNS"	LABEL
Label-sm	Family: Satoshi Weight: 500 Size: 10px Line Height: 14 Letter-spacing: 0.8 UPPERCASE Use: Bottom Nav tab labels, chip labels (STUDY MODE, HOME), time blocks in task cards	LABEL
Button	Family: Satoshi Weight: 500 Size: 14px Line Height: 20 Letter Spacing: 0 Use: Button text on Primary and Secondary buttons	Button Text
caption	Family: Satoshi Weight: 400 Size: 11px Line Height: 16 Letter Spacing: 0 Use: Timestamps, fine print, accessibility labels	Link Text

Full type scale — XL title (32px) down to caption (11px)

Logo — The Floating Dot

The mark has two elements: an arch and a coral dot floating above it. The arch represents the pressure pushing down on the student. The dot, suspended above the arch's apex, represents the student's mind — the headroom. The mark works at favicon size, scales cleanly across light and dark modes, and reads as a brand symbol before it reads as a wordmark.

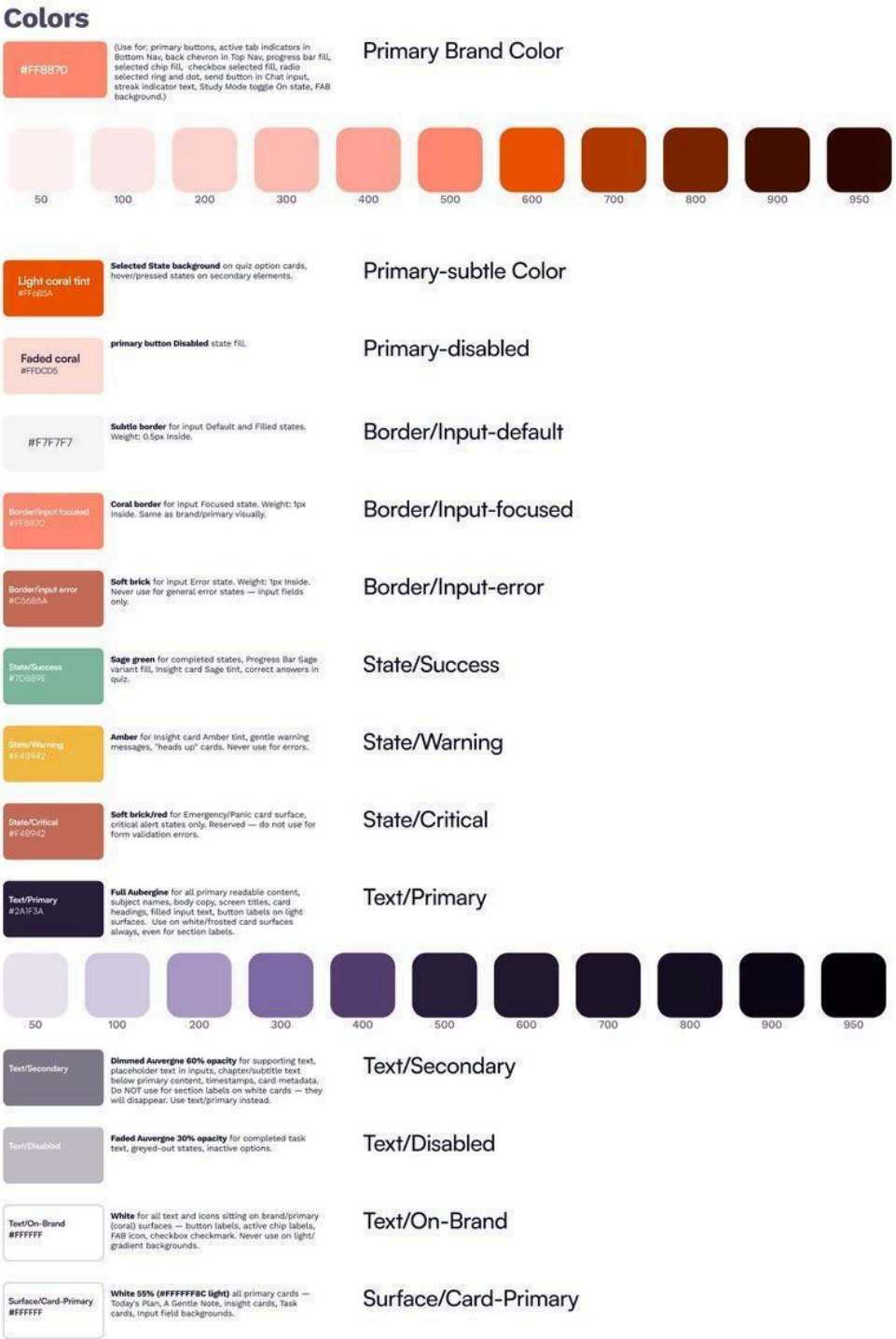
The Floating Dot — Headroom logomark



The Floating Dot logo — arch + coral dot

Colour system

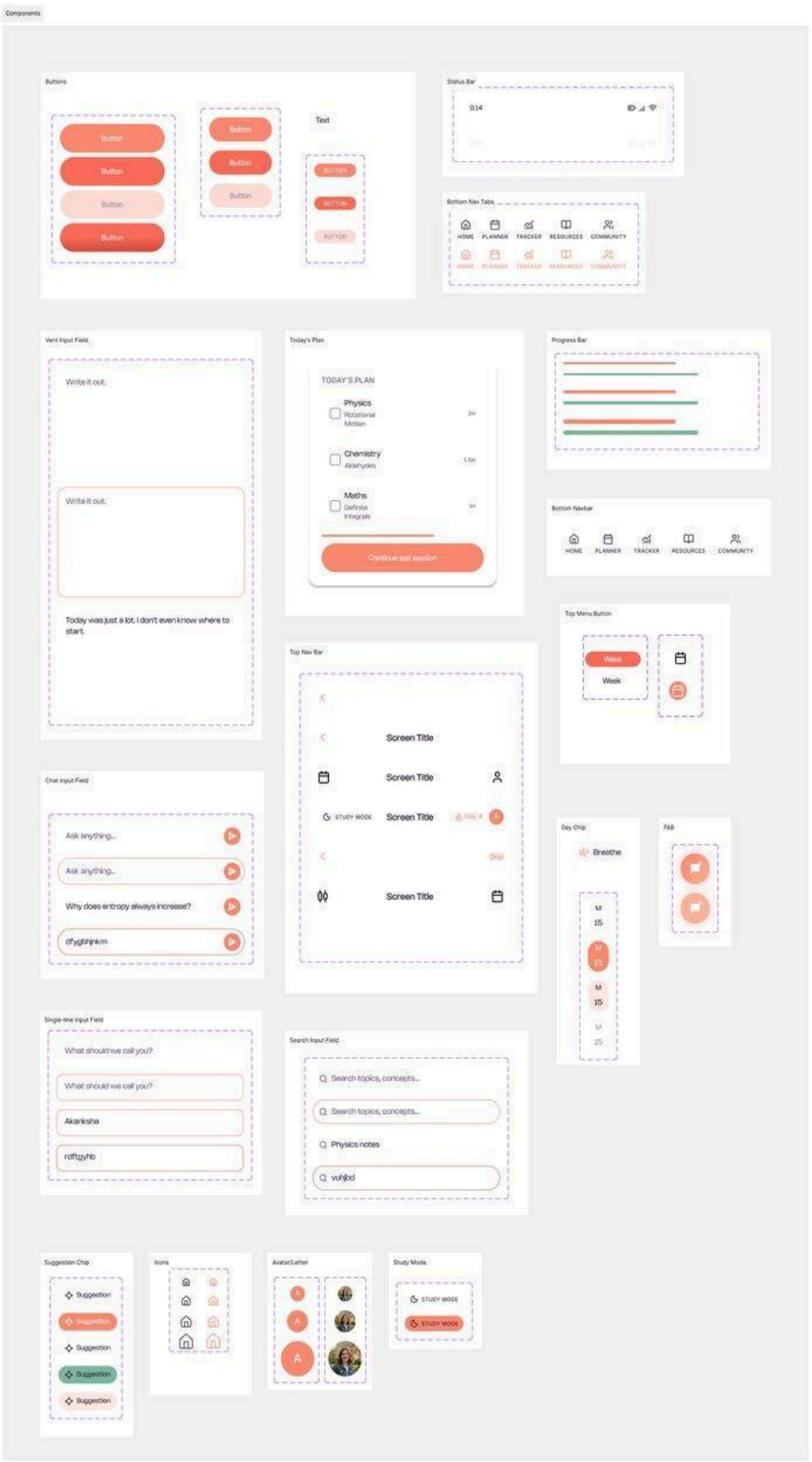
The primary brand colour is coral (#FF8870) — warm enough to feel welcoming, distinct enough to anchor brand recognition, restrained enough to avoid signalling alarm or urgency. Deep aubergine (#2A1F3A) carries all primary text. A sage green (#7DB89E) is reserved for completed states. The palette runs across twelve coral tints and twelve aubergine tints for system-wide consistency.



Full colour system — coral, aubergine, semantic states

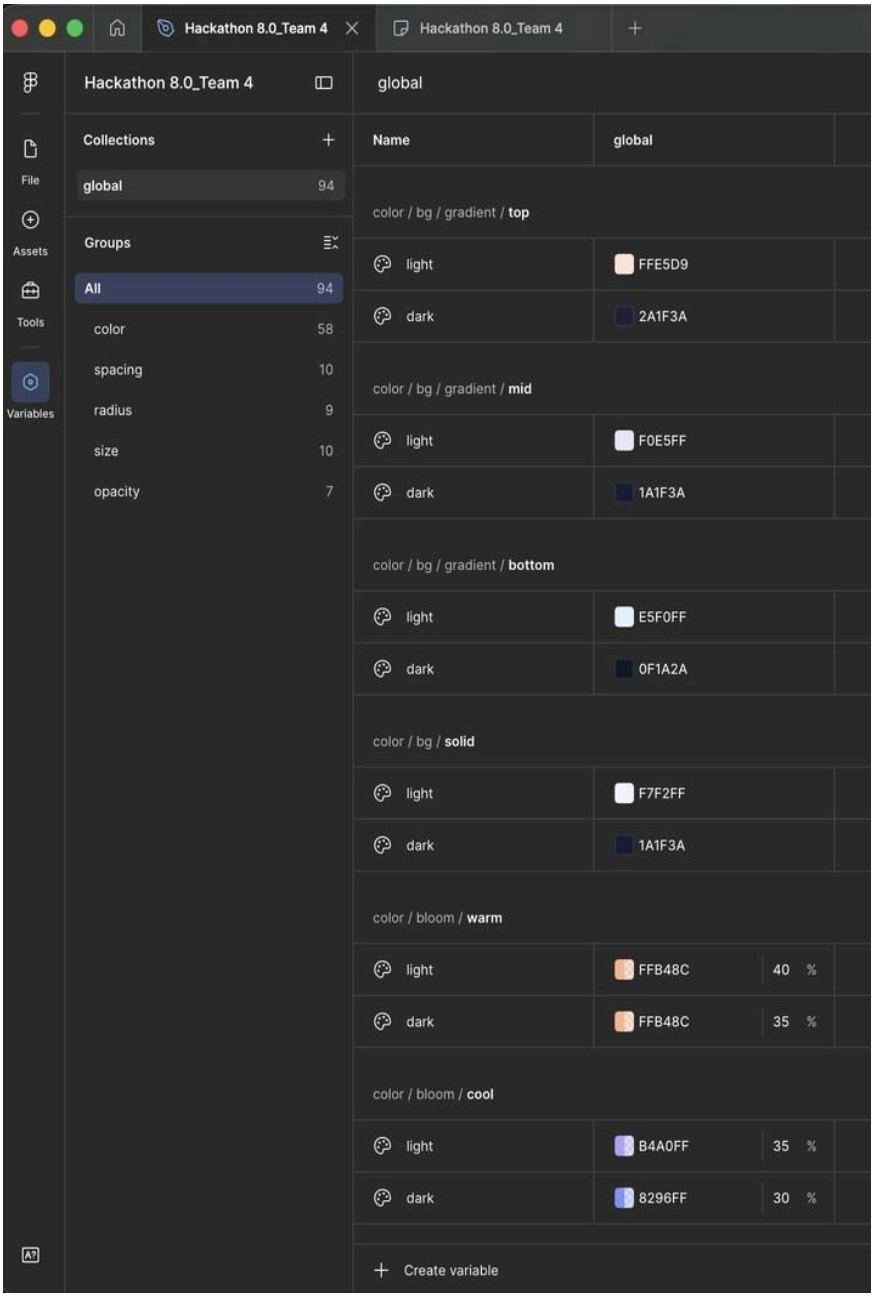
Components — 122 tokens

The design system shipped with 122 tokens covering colour, typography, spacing, radius, shadow, and motion, plus a full Figma component library — buttons in four variants, input fields with focused and error states, frosted cards, chips, top and bottom navigation, FABs, suggestion chips, and the day chip used in the Tracker. Token Studio was the source of truth across all design files.

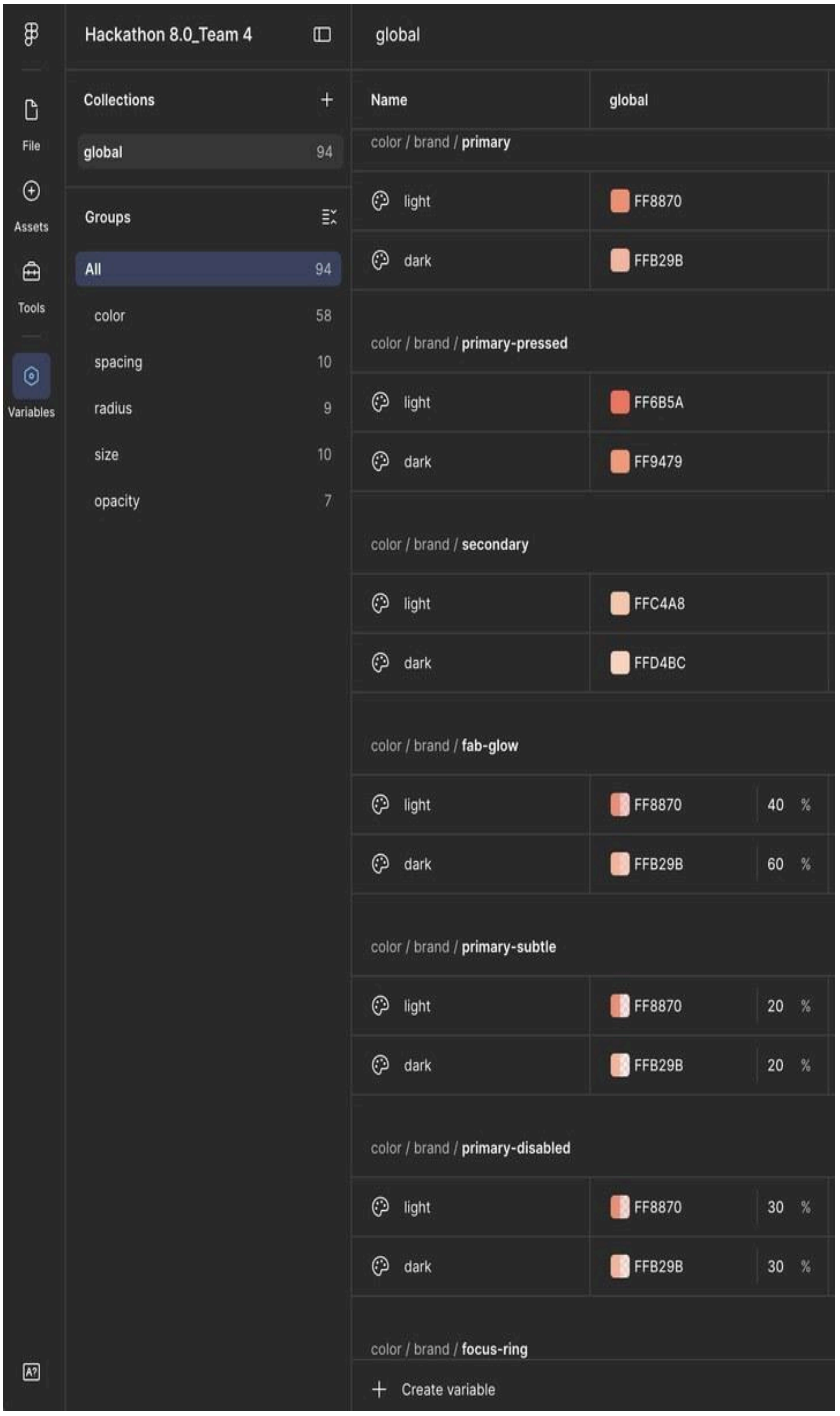


Component library — buttons, inputs, nav, cards, chips, FAB

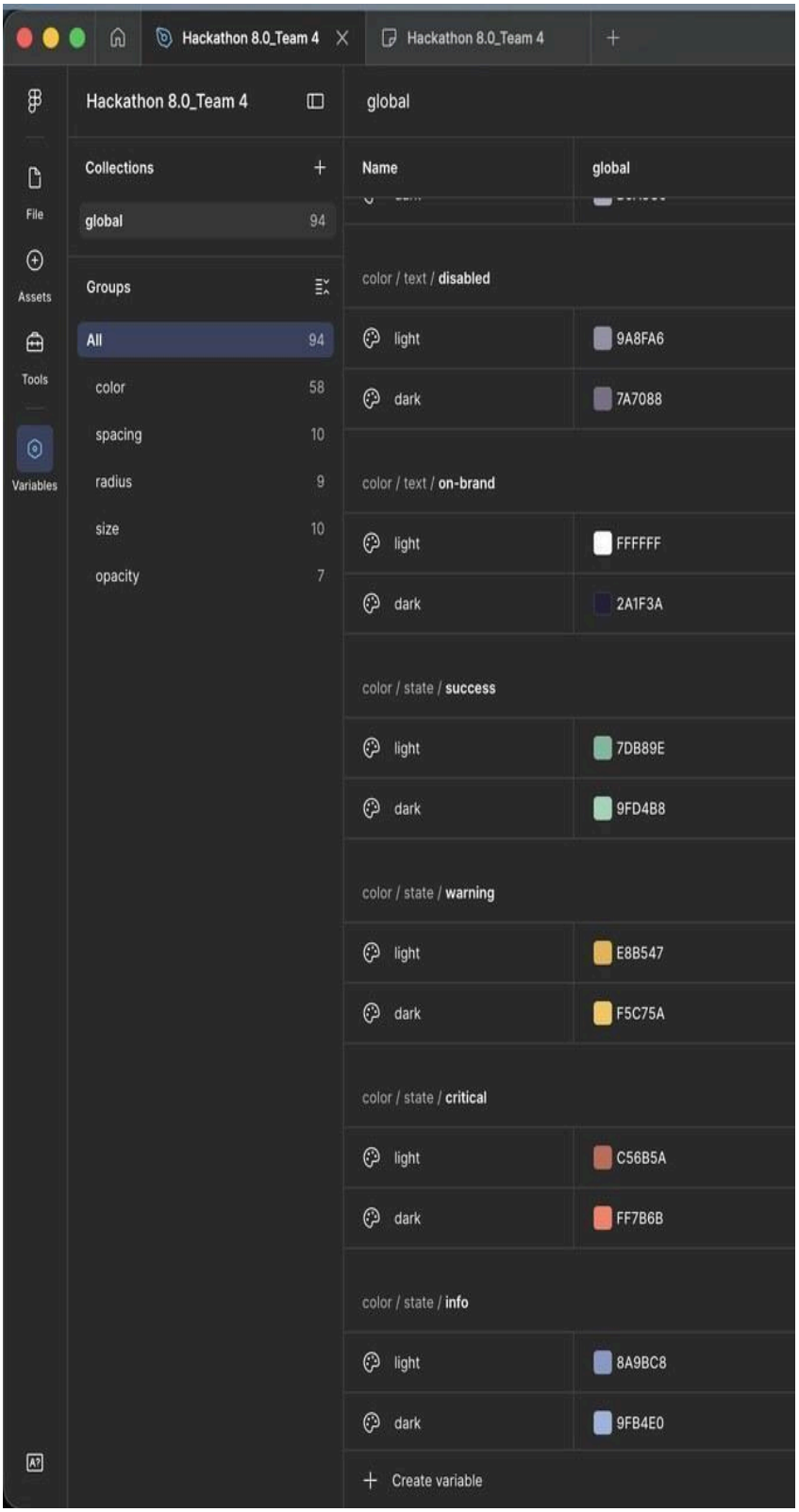
Design tokens — Figma Variables



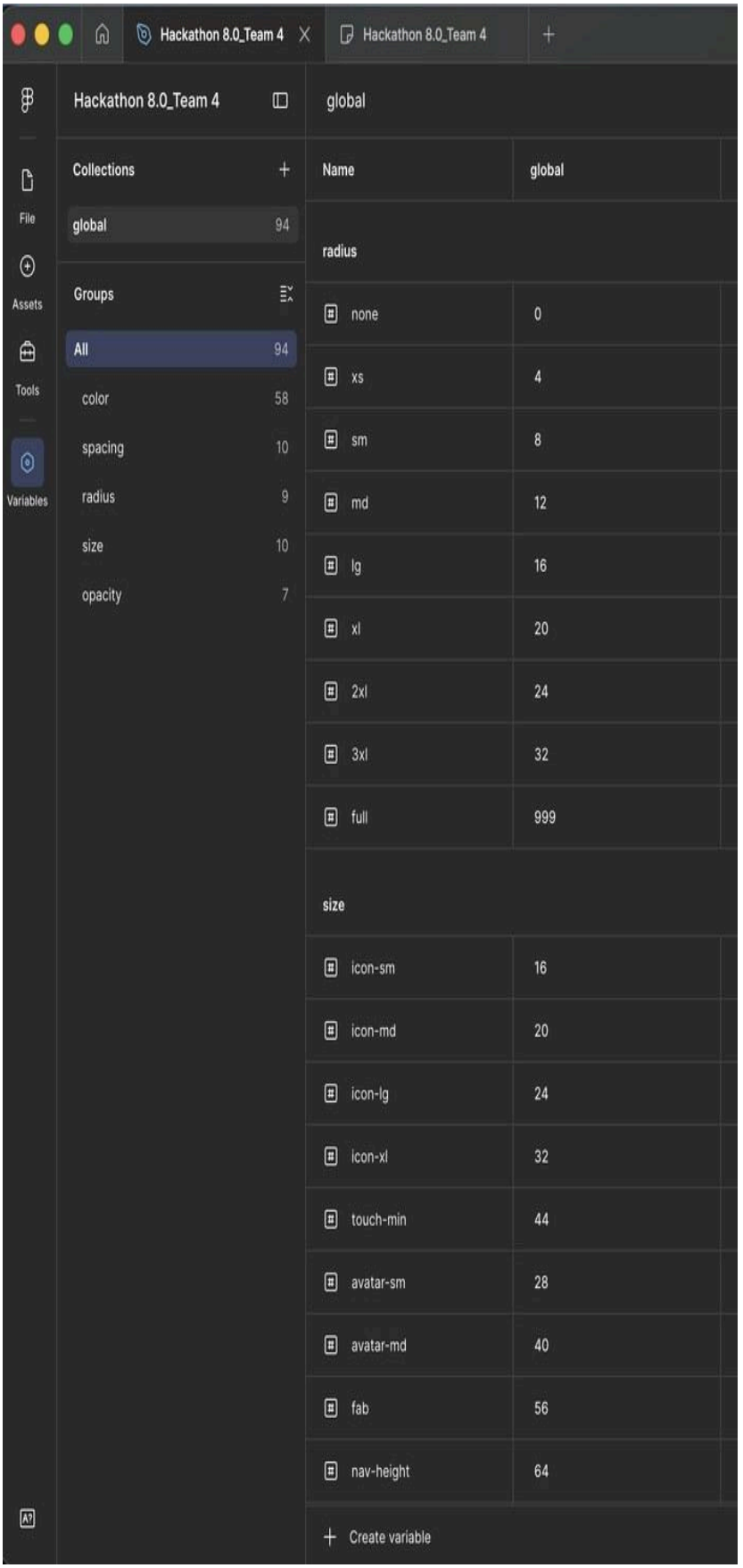
Figma variables — background gradients and bloom tints



Figma variables — brand primary, secondary, and state tokens



Figma variables — text tokens and semantic state colors



Figma variables — radius scale and size tokens (94 total)

SECTION 07

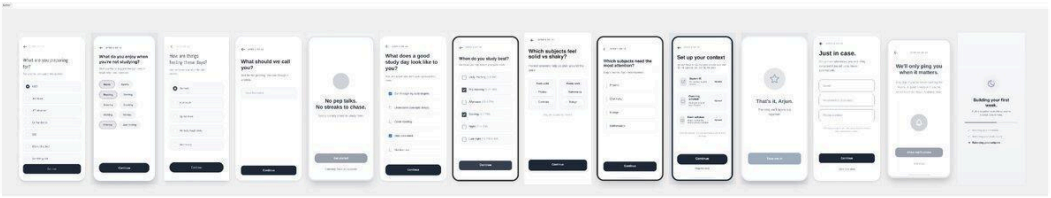
Prototype

Prototyping built on the low-fidelity wireframes from the architecture phase. Mid-fidelity was the next step, where structure met real copy and component density. Prototyping ran in two parallel tracks — a Figma prototype that carried the visual design and the full product narrative, and a working HTML prototype that made the AI features genuinely testable. Both were needed for different reasons.

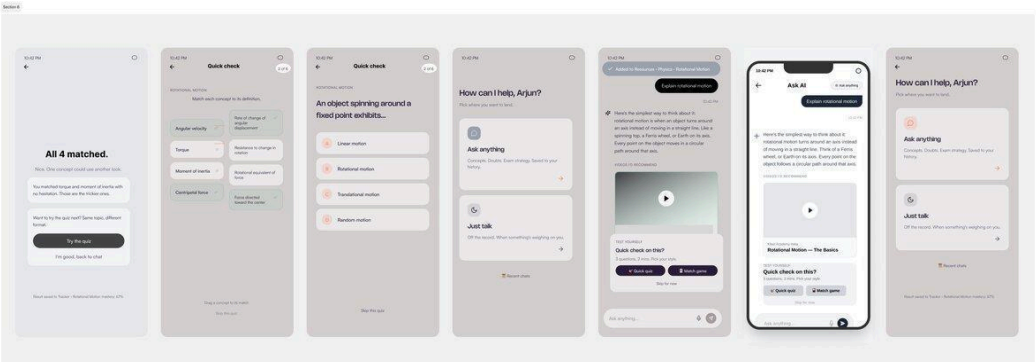
> **Figma Design File** Open in Figma ↗
>
> **Interactive Prototype** Open Prototype ↗
>
> **FigJam Working Board** Open FigJam ↗

Mid-fidelity in Figma

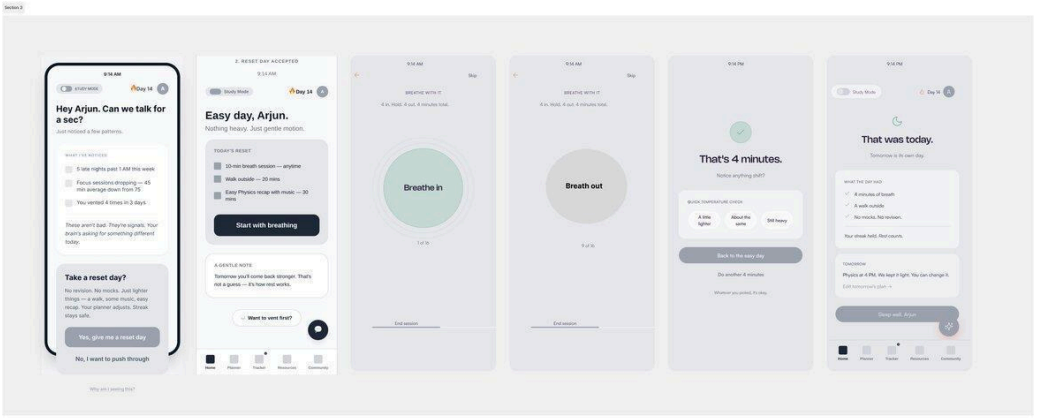
Once lo-fi structure was approved, we moved to mid-fidelity for eight core surfaces — Onboarding, Daily Study Flow, Planner, Tracker, Ask AI, Buddy, Vent, and the Burnout Reset flow. Mid-fidelity was where we introduced layout, hierarchy, real copy, and component density — but still in greyscale, deliberately. The team was still resolving the visual system in parallel, and we did not want layout decisions to be muddied by colour and brand choices that hadn't fully landed. Mid-fi was the version we used for internal team critique and the first round of usability testing.



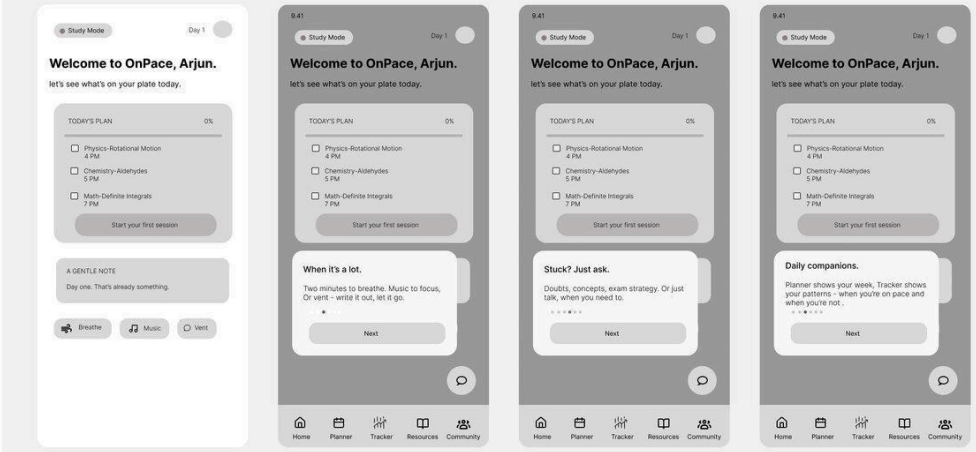
Mid-fi — Onboarding flow



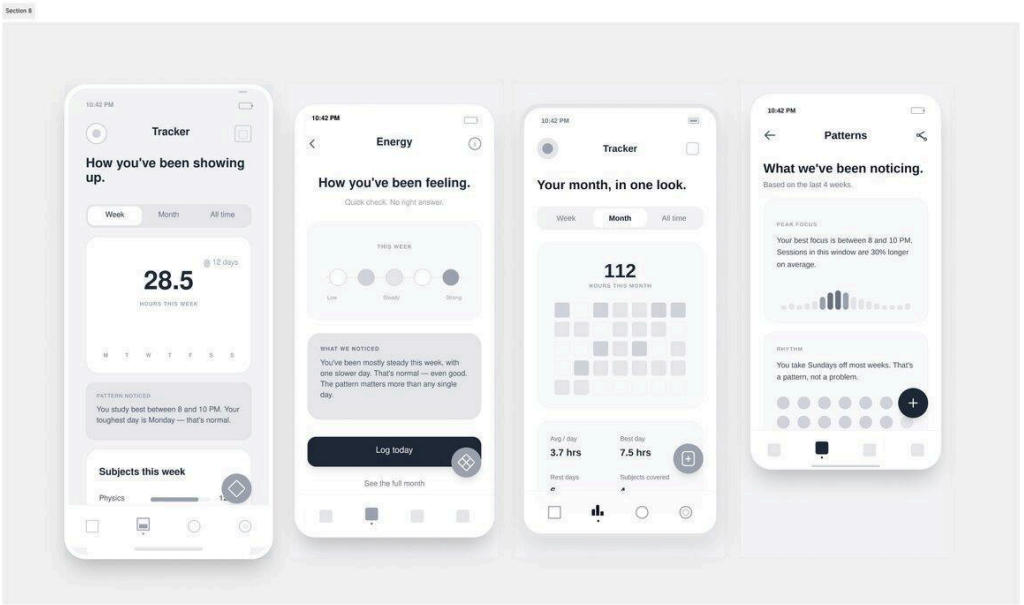
Mid-fi — Ask AI screens



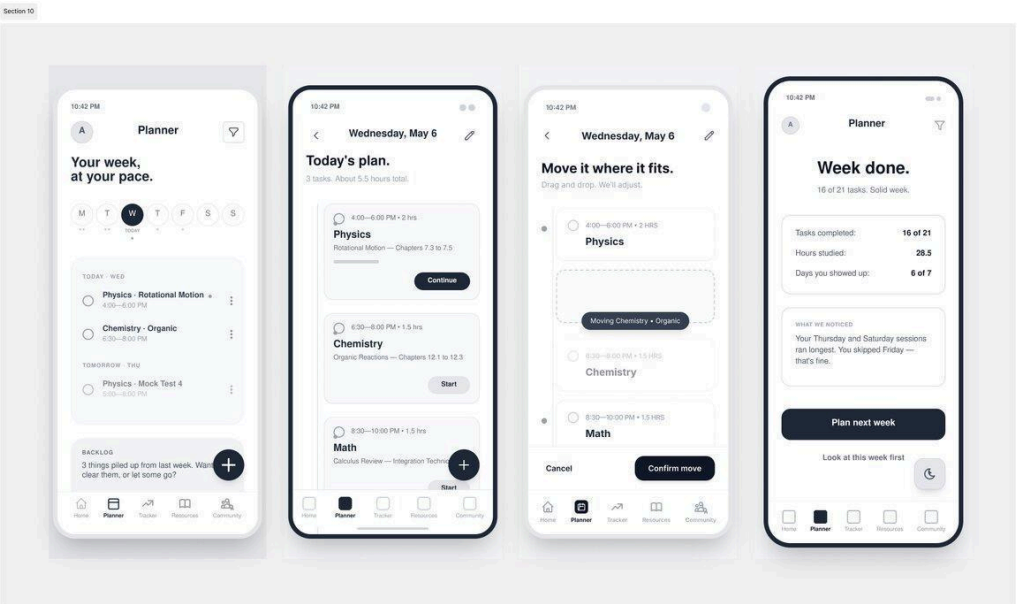
Mid-fi — Burnout Reset flow



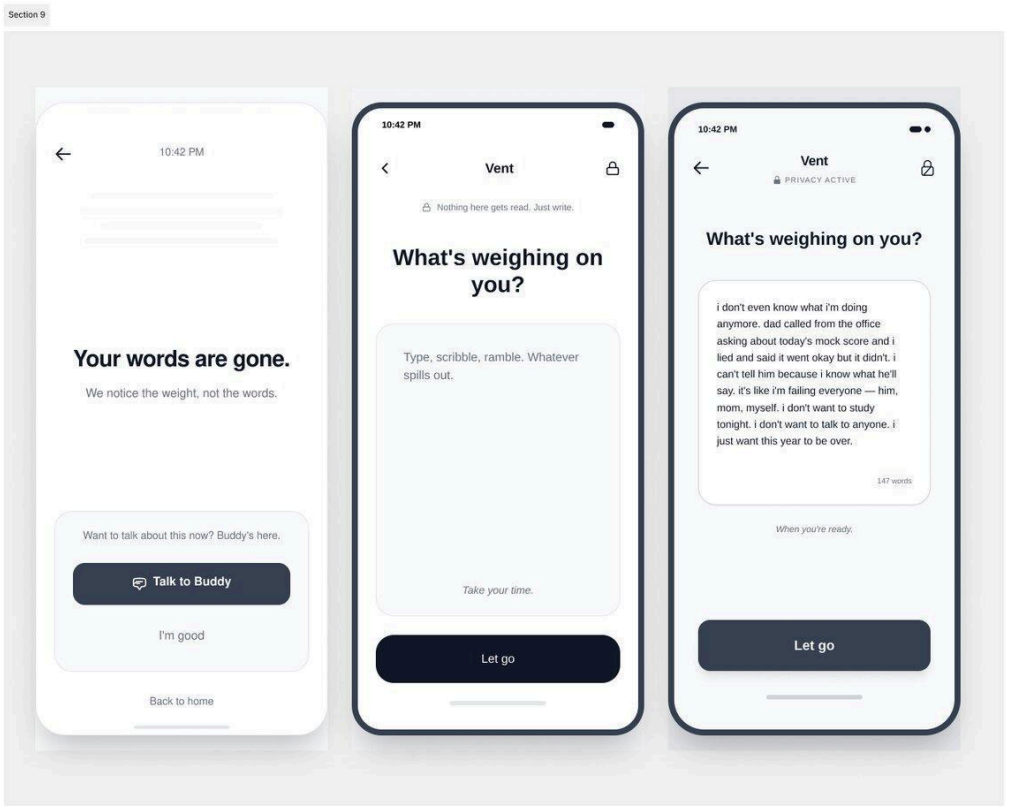
Mid-fi — Feature popup / onboarding tour



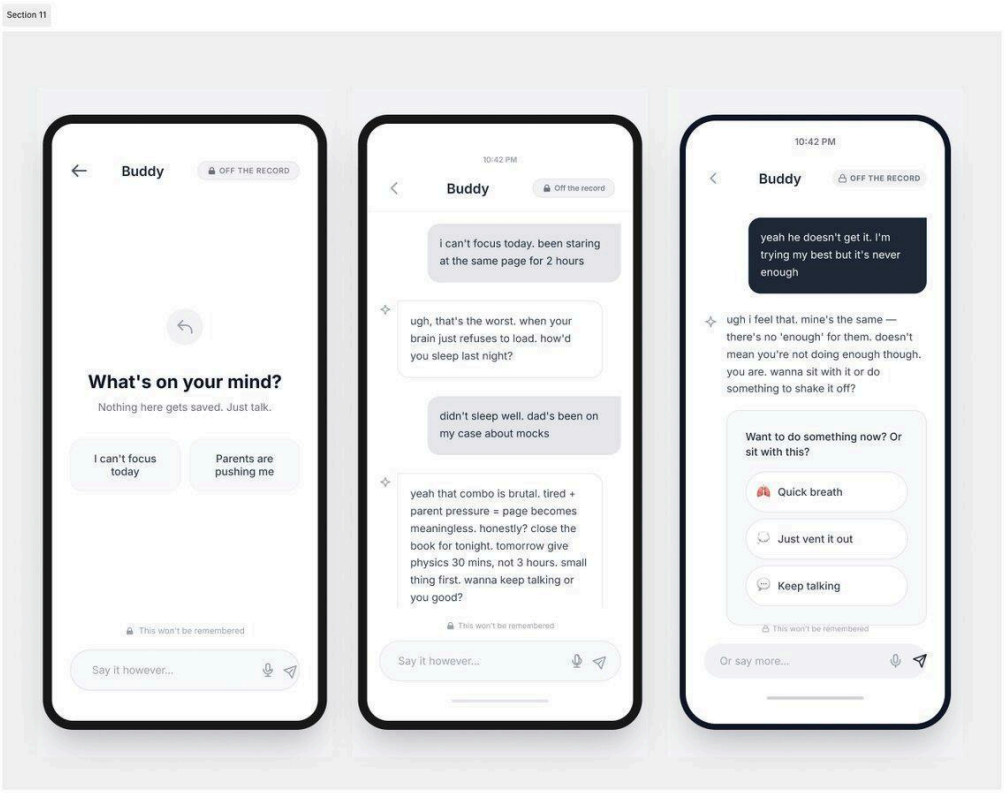
Mid-fi — Tracker



Mid-fi — Planner



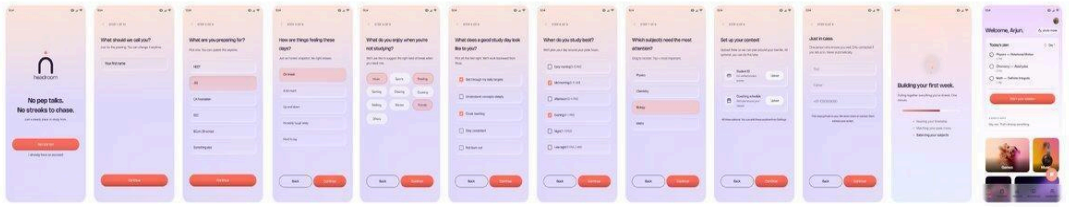
Mid-fi — Vent space



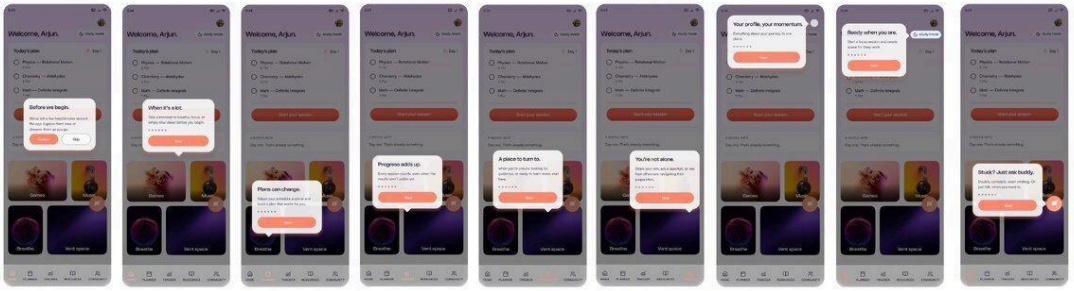
Mid-fi — Buddy chat

High-fidelity

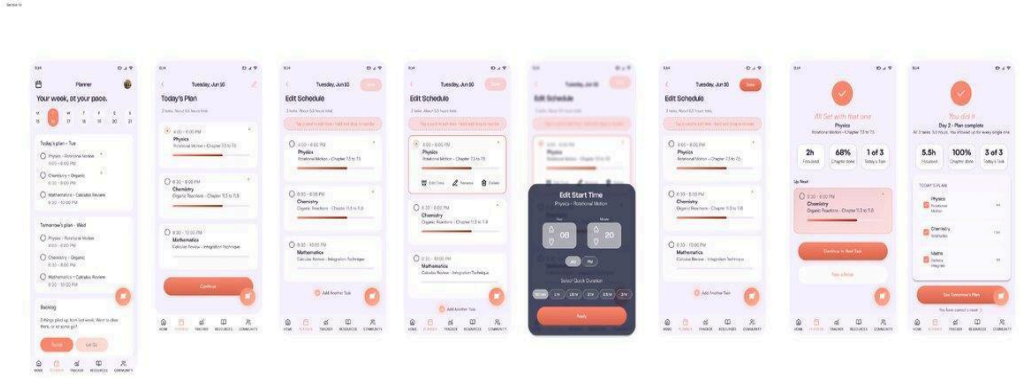
High-fidelity rolled out surface by surface as the Glass Sanctuary system stabilised. Onboarding and the Feature Popup were the first fully resolved screens, followed by Home, Buddy, Vent, Burnout Reset, Ask AI, Planner, and Tracker. Each screen was built using only design system components — no off-grid one-offs. Token Studio drove every colour, type, and spacing value.



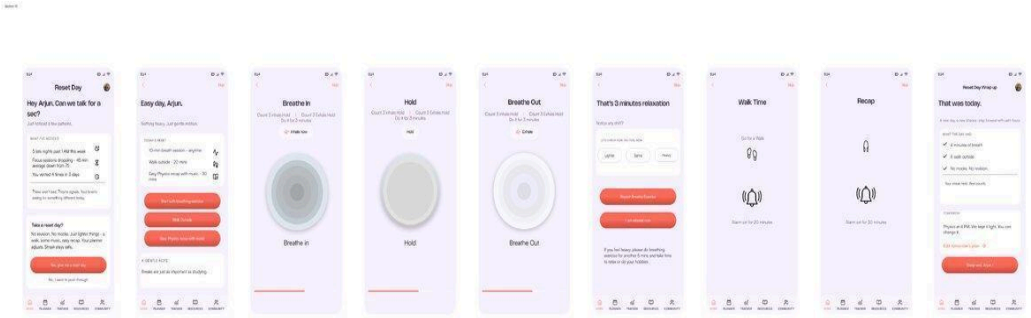
Hi-fi — Onboarding flow



Hi-fi — Feature popup tour



Hi-fi — Planner screens



Hi-fi — Burnout Reset flow

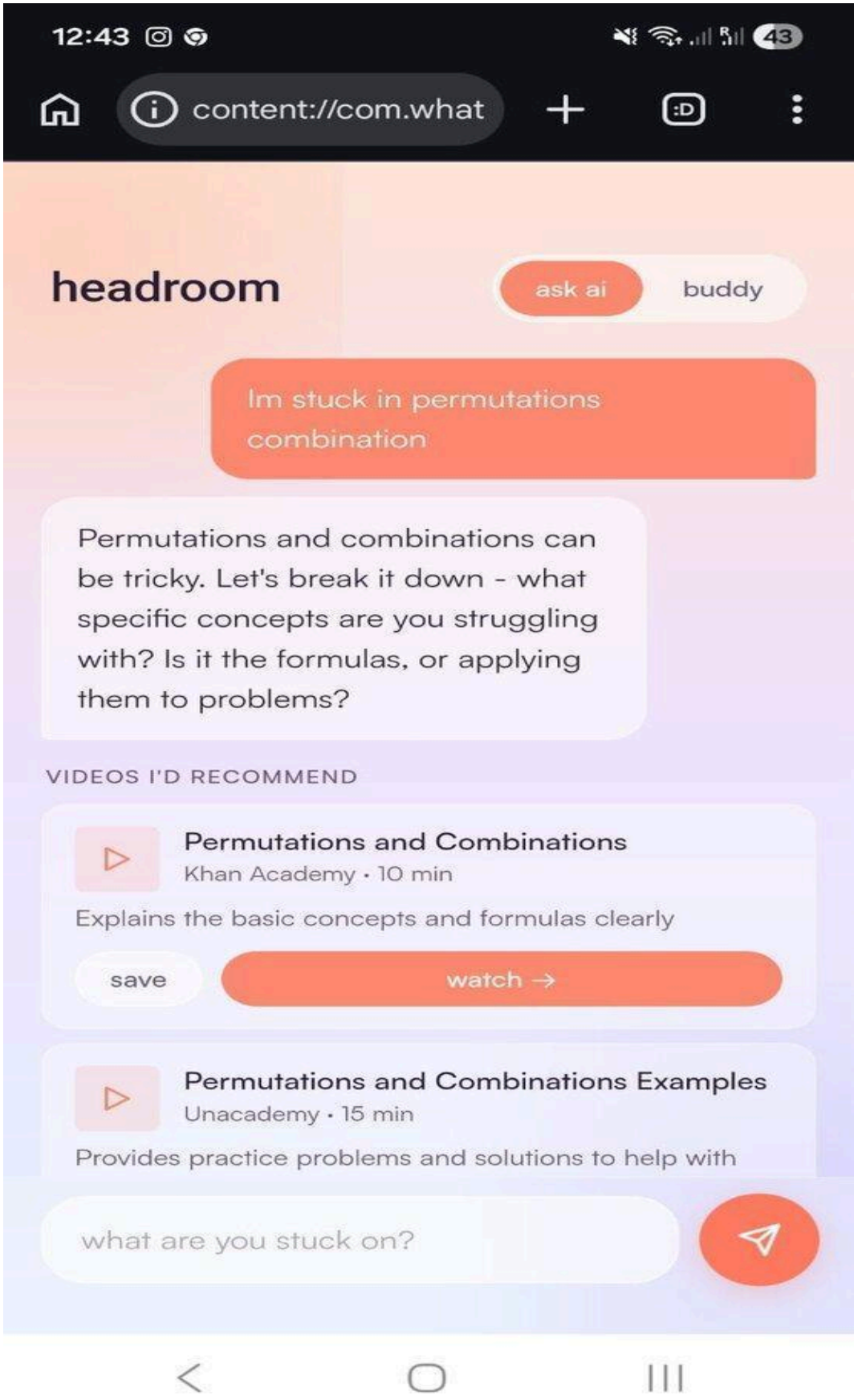
HTML prototype — the real AI test

A Figma prototype can fake an AI conversation. It cannot test whether the conversation is actually useful, safe, or in the right voice. Two of our most critical features — Ask AI and Buddy — depend entirely on the AI behaving correctly. So we built a working HTML prototype, hosted on Netlify, connected to Groq's API running Llama 3.3 70B. Both Ask AI (older-sibling tutoring voice) and Buddy (same-age peer support voice) were wired up to actual prompts.

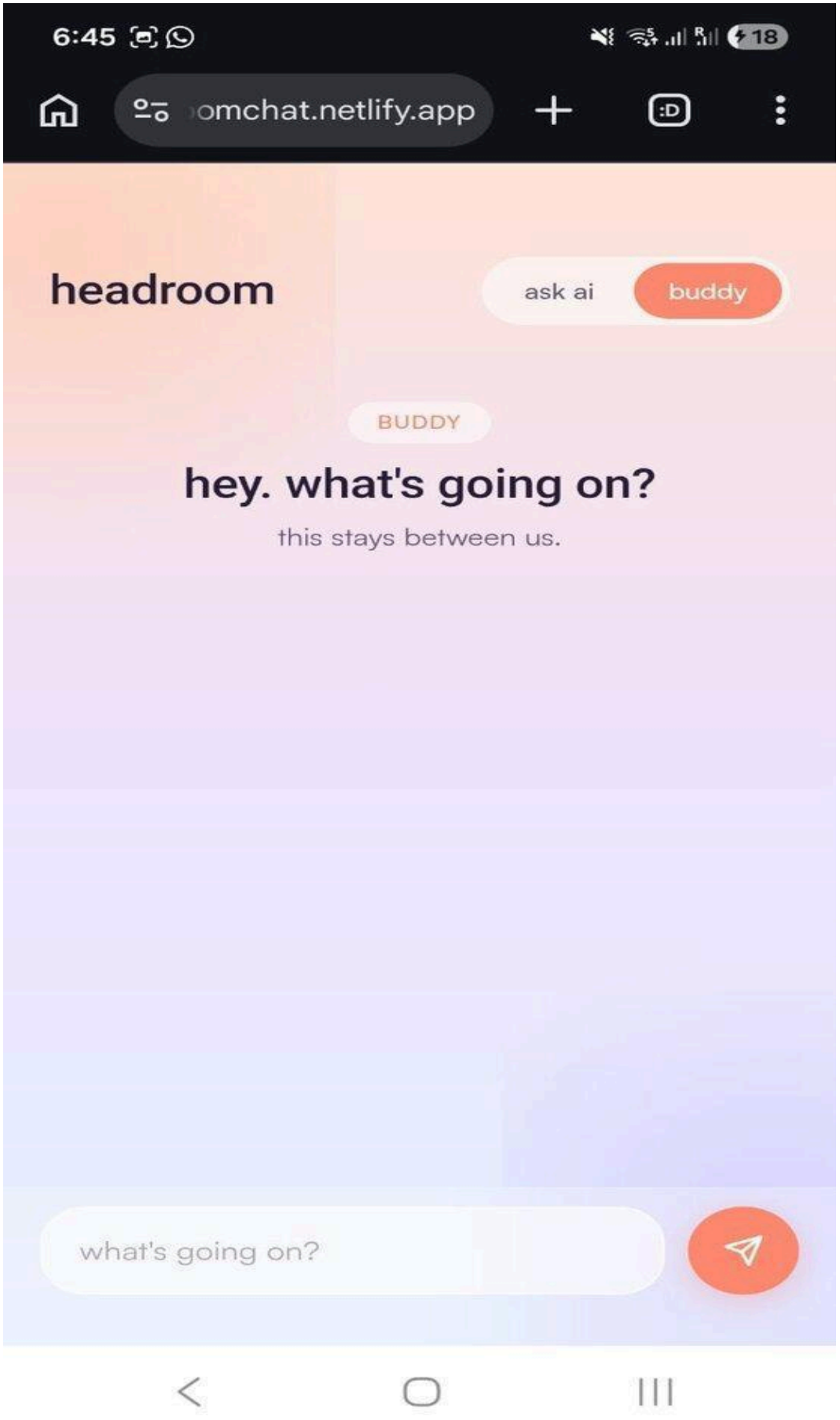
Seven rounds of prompt refinement followed. The first version of Buddy slipped into therapeutic mirroring — patterns we had explicitly written research findings against. By round seven, both prompts were consistently producing the voice we had designed for.



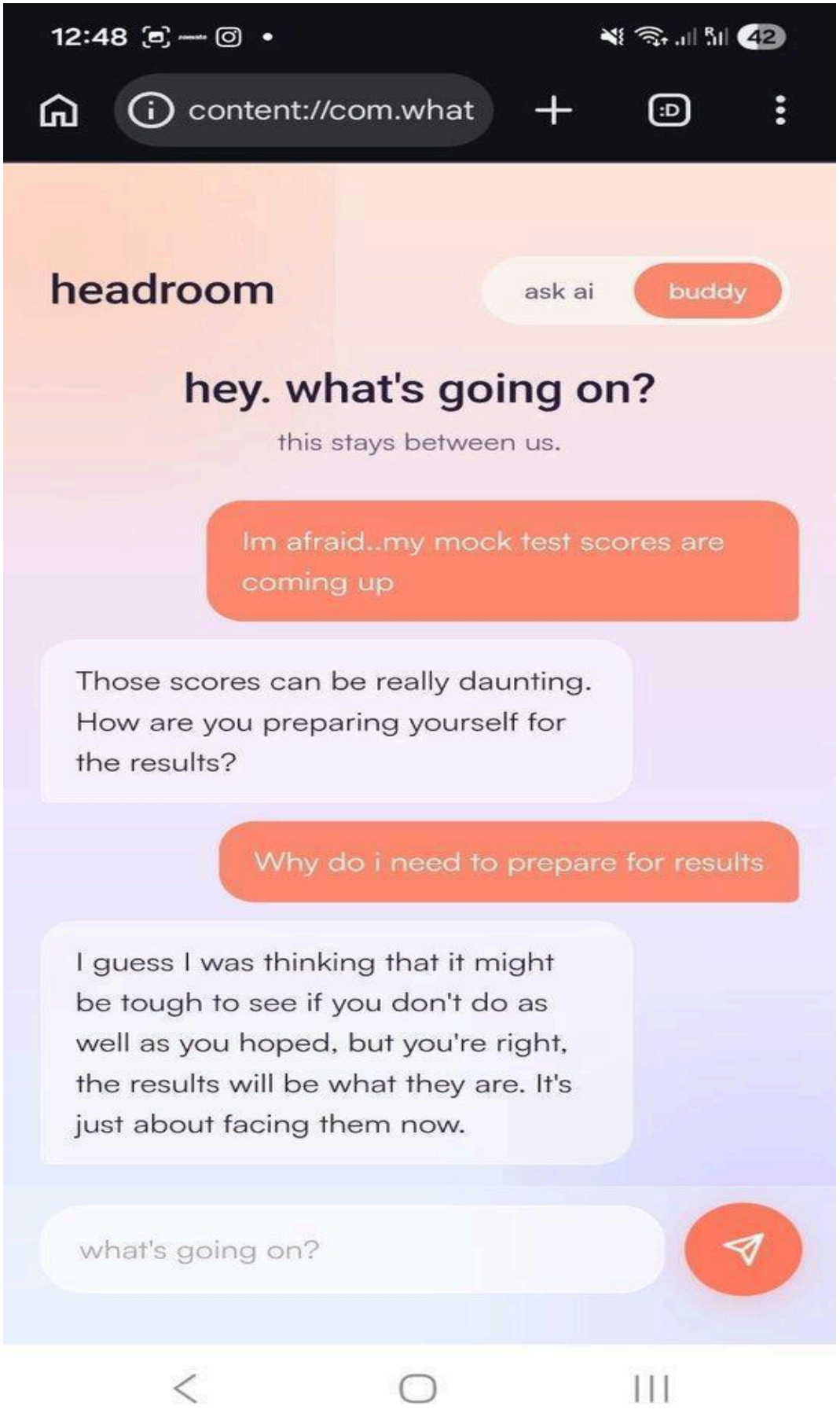
HTML prototype — Ask AI entry



HTML prototype — Ask AI conversation (permutations & combinations)



HTML prototype — Buddy entry ("hey. what's going on?")



HTML prototype — Buddy conversation (mock test anxiety)

SECTION 08

Testing & Iteration

Testing happened in two formats — a structured usability study run on Maze, and informal in-team task walkthroughs that ran continuously across iteration cycles. Both formats served different purposes. The Maze study generated quantitative task completion data and timed friction signals. The in-team walkthroughs surfaced emotional reactions and language issues that quantitative data couldn't capture.

Study setup

We recruited participants from our research pool — students currently preparing for competitive exams, fitting our 15–19 age window. The full task set covered six tasks spanning the highest-priority flows: onboarding, daily study mode, Ask AI, planner editing, tracker, and the Burnout Reset surface. Ten participants completed the full task set. Five to six participants dropped off mid-test — a drop-off rate we documented as a finding in itself: any future iteration would shorten the task set or break it into two shorter sessions.

Task-by-task results

Task 1 — Onboarding + Home Screen (94% completion, 31.6% misclick rate, 3.5/5 ease): Most participants correctly identified the app as study-support oriented. Key friction: the step count itself, and the colours/layout of initial screens feeling unfamiliar. One participant noted "colours and lines were confusing." Setup ease averaged 3.5/5 across 11 respondents.

Task 2 — Daily Study Flow: Key friction was starting a session — unclear whether tapping Study Mode initiated a specific subject session. "Not knowing if that started physics study time" came up across multiple sessions. Highest-friction task in the set.

Task 3 — Ask AI (89% completion): The floating button (FAB) was not immediately understood before tapping. After tapping, the Ask AI surface was intuitive and response quality was rated positively.

Task 4 — Planner Edit: High completion. Majority said editing felt straightforward. "Does this match the way you actually think about your study week?" — most said yes.

Task 5 — Tracker (free explore): Participants responded positively to observational language — no judgement framing. Multiple noted it helps them track studies. Most useful element cited: the tracker itself.

Task 6 — Burnout Reset (100% completion): The nudge card landed as intended — "looking out for me" was the dominant response (5/7). Several felt "easy and relaxed" after seeing it.

Top friction points

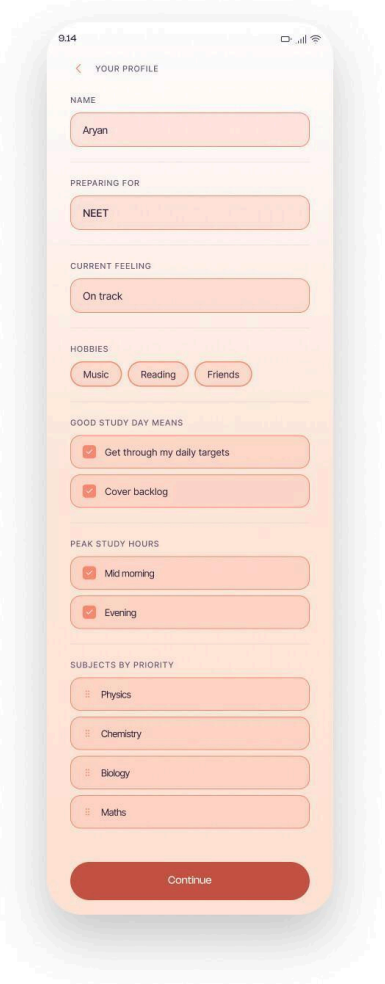
- Starting a study session — unclear whether Study Mode initiated a specific subject session
- Onboarding misclick rate (31.6%) — highest of any task, indicating navigation ambiguity
- FAB recognition — Ask AI entry not immediately understood before interaction
- Subject selector logic — biology appearing for a JEE aspirant
- Feature tour density — multiple popup messages on first home screen felt overwhelming

Top validation signals

- 100% Burnout Reset completion with dominant "looking out for me" sentiment
- Planner editing rated straightforward — core study structure is usable
- Ask AI response quality received positive feedback once reached

- Tracker's observational language resonated
- Product intent clearly understood — described as study-support, not therapy-facing

Iterations made from testing



Onboarding summary screen — "Your Profile" — changes made post-testing iteration.

Study Mode entry friction → Added explicit subject confirmation step so system state is visible before a session begins.

FAB visibility → Added brief label hint on first-time Ask AI access. Label disappears after first interaction.

Feature popup density → Reduced home screen tour cards from 8 to 5.

Subject selector logic → Exam-type mapping tightened so JEE profiles don't surface Biology as default.

Confidence statement

> We know with conviction: the emotional framing works, Burnout Reset lands as intended, Planner structure is usable, and the product's intent reads clearly to our target population. What still needs further testing post-hackathon: the full Ask AI conversation flow across varied question types, the Vent write-and-release experience in a real late-night context, and the long-term pattern recognition in the Tracker.

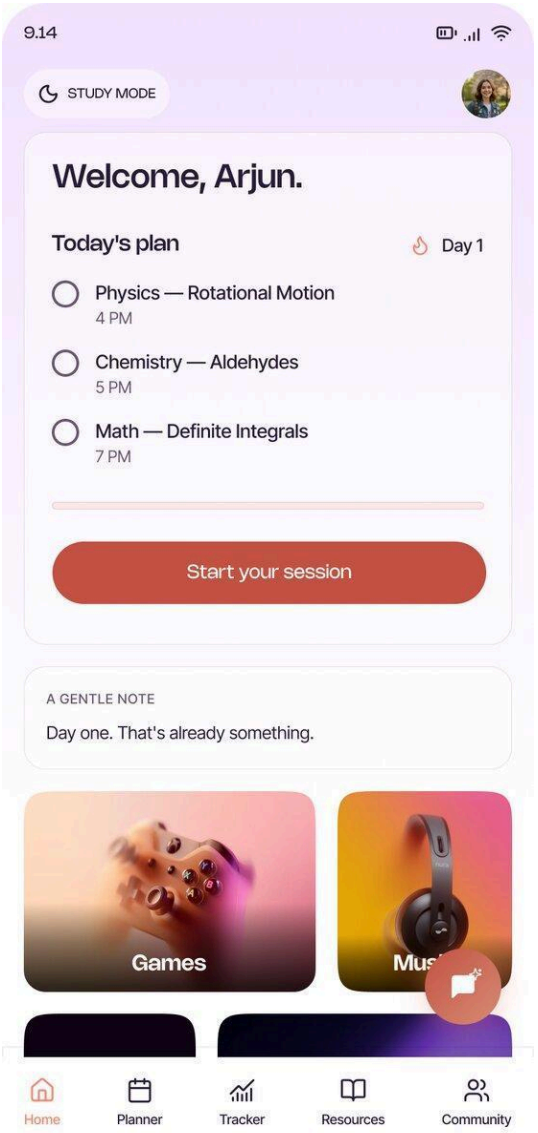
SECTION 09

Final Solution

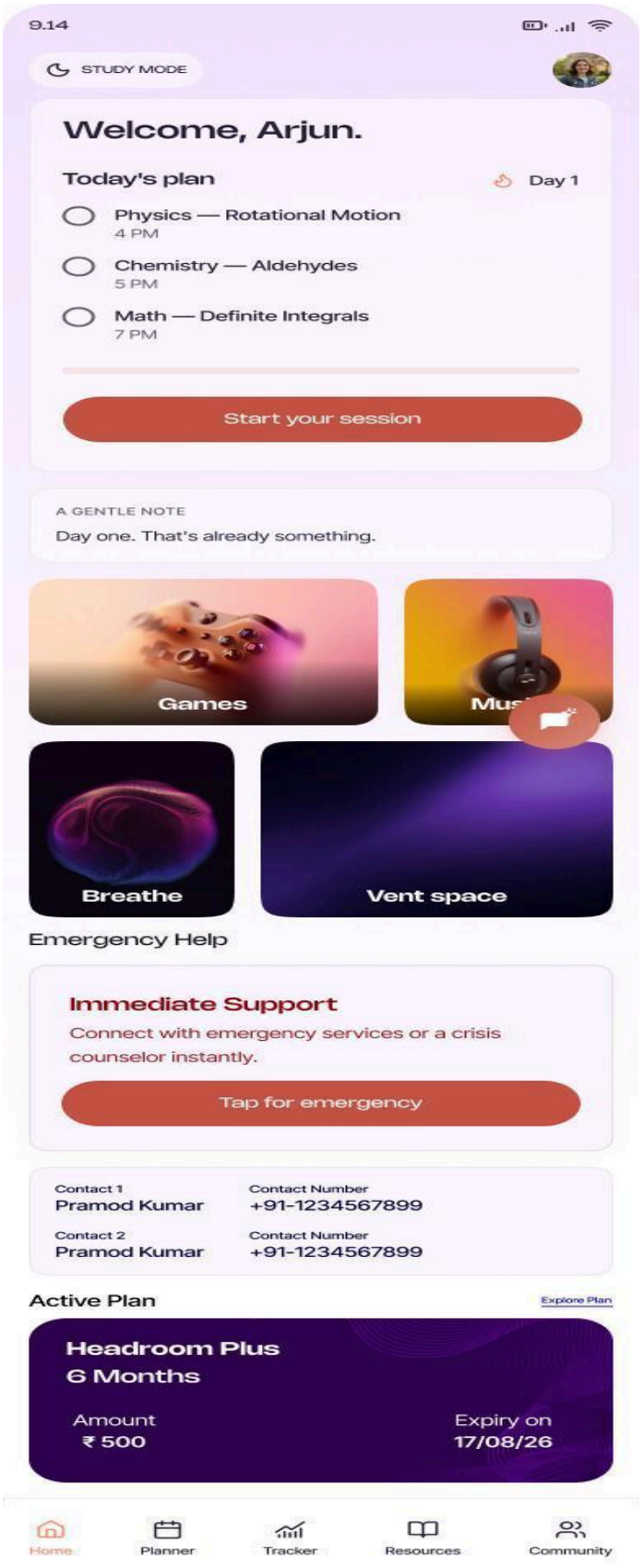
The final product is built on a simple architecture: a visible layer that performs as a study app, an embedded layer that performs as emotional support, and a safety layer that's always free.

The visible layer — what the product looks like

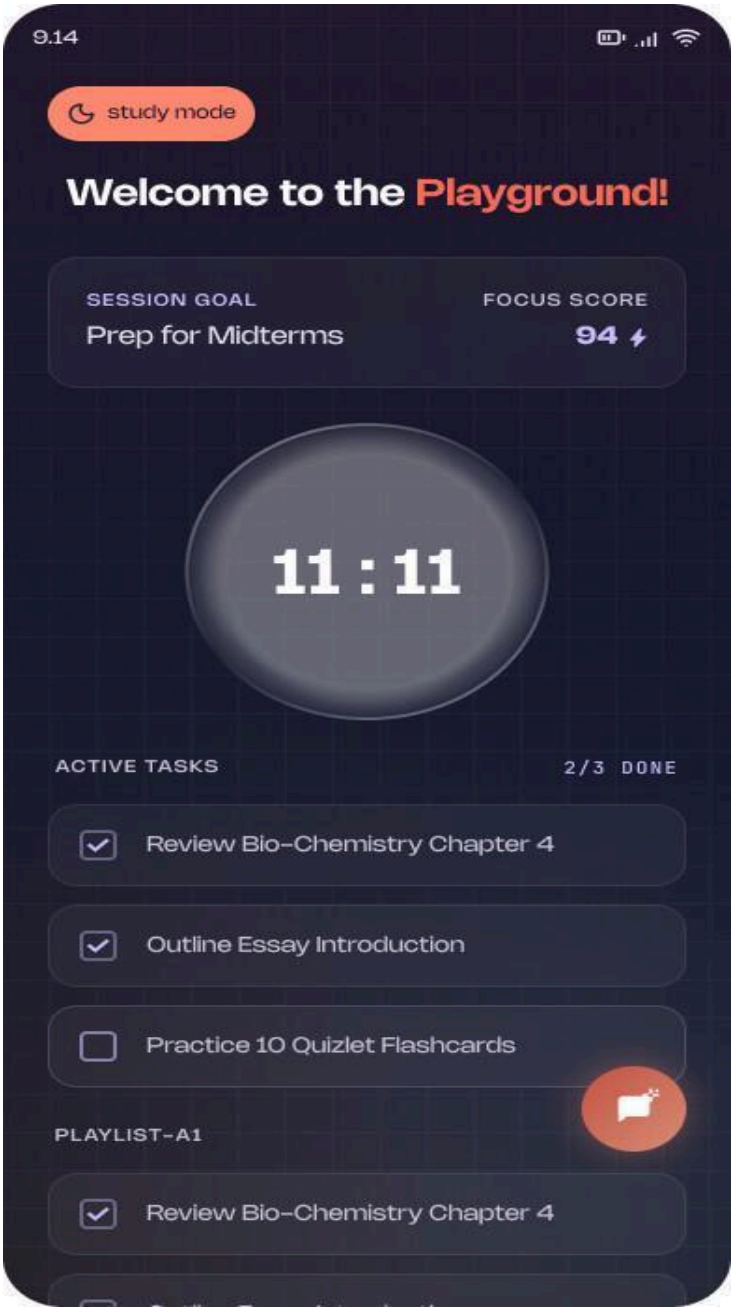
The Home screen leads with Today's Plan — a clean breakdown of subjects and study blocks for the day, with a single progress indicator. Below that, A Gentle Note offers a soft framing of where the student is in their week. Below that, Patterns surfaces lightweight observations the system has noticed across recent days. No mood check-in. No anxiety scale.



Home screen — Today's Plan, Gentle Note, embedded quick actions

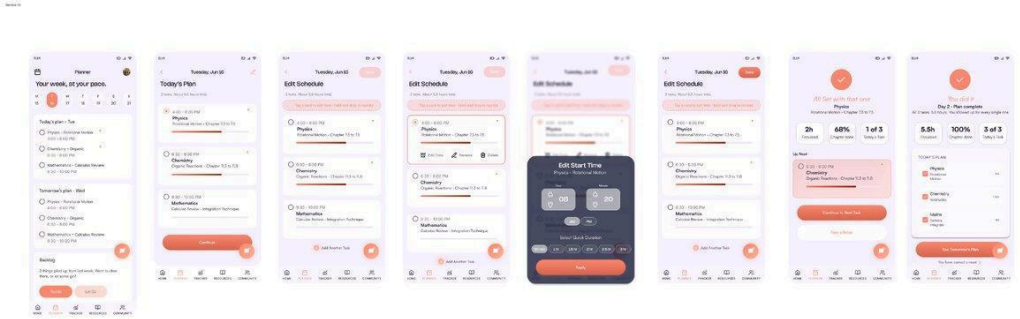


Home screen — Today's plan, Gentle Note, quick-access emotional layer.

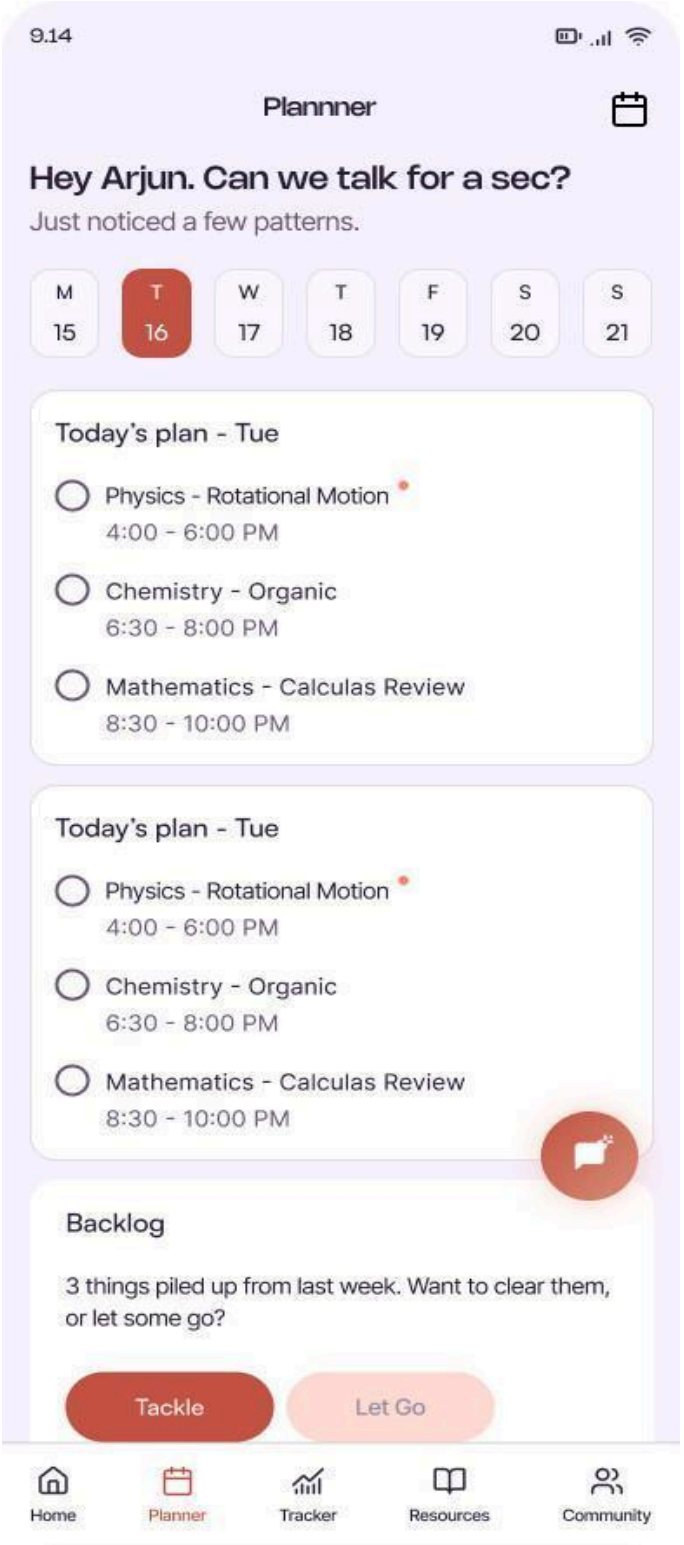


Study Mode — the Playground. Focus timer, active tasks, momentum streak.

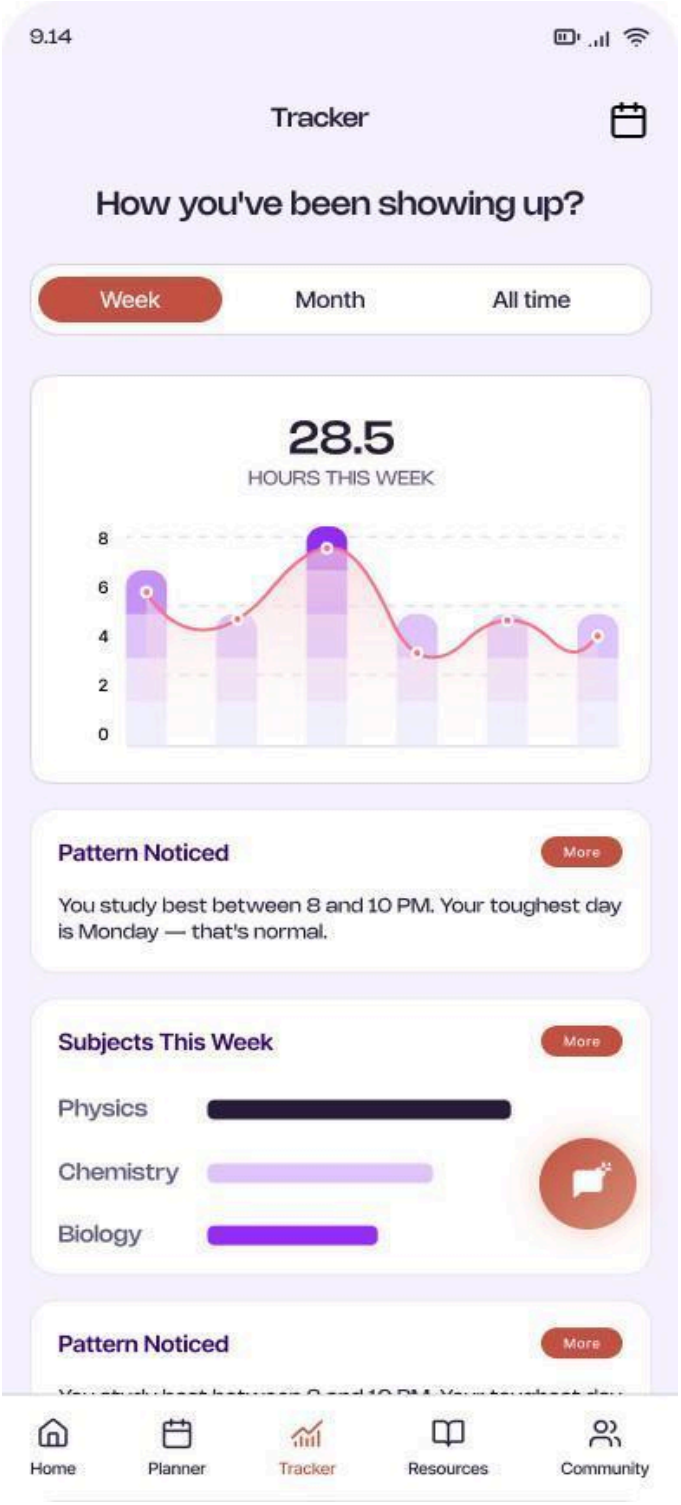
The Planner gives the student a weekly view that bends around emotional reality. Backlog from missed days surfaces in a Backlog card with a single prompt: clear it, or let some of it go.



Planner — weekly view, backlog card, drag-to-reschedule



Planner — weekly view, backlog rescue, Tackle or Let Go.



Tracker — How you've been showing up. Study patterns without comparison.

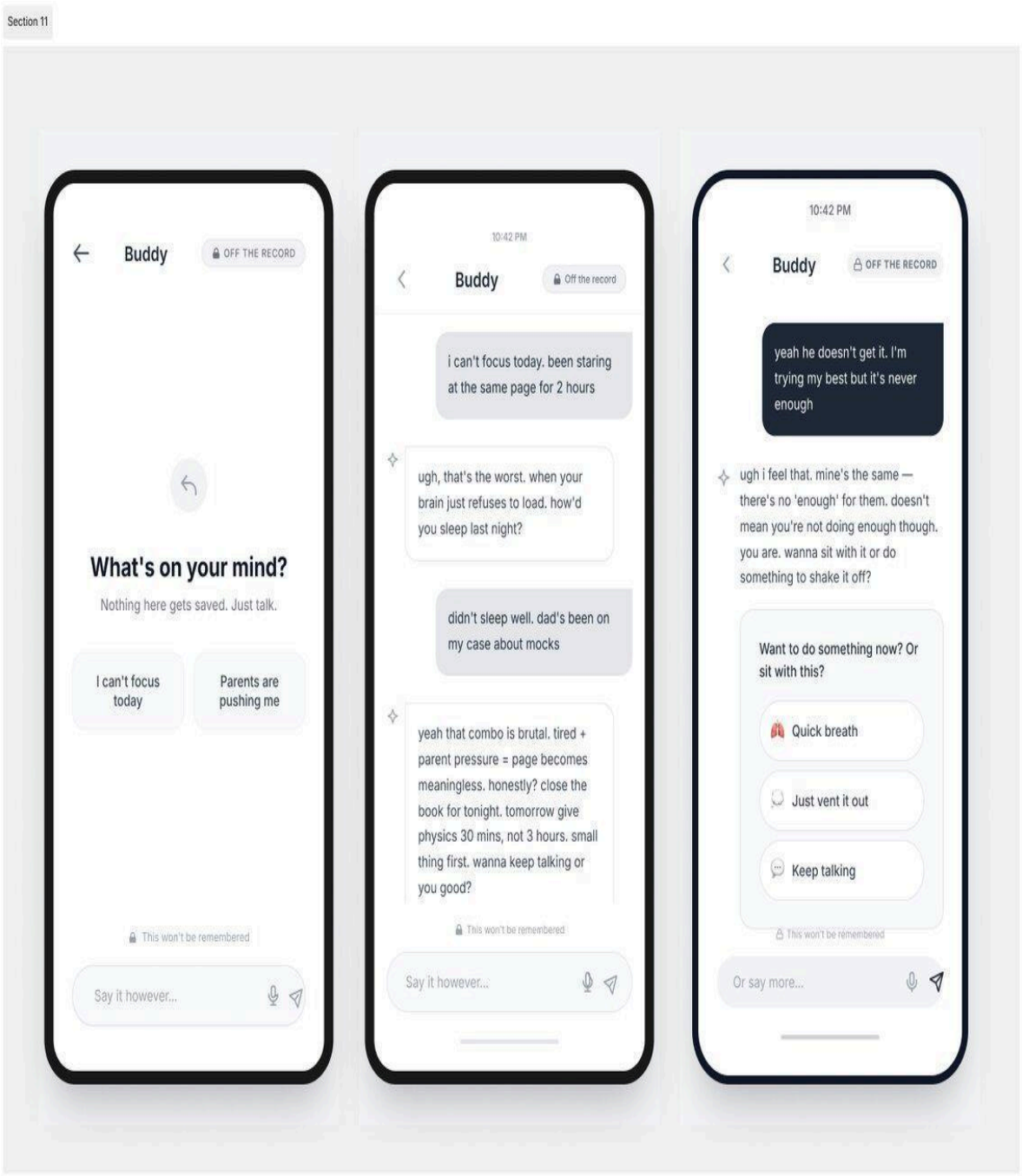
The embedded layer — hidden inside

Ask AI is the doubt-clearing surface. The student picks a subject, asks a question, and gets an explanation calibrated to their level — older-sibling tone, not textbook tone. Inside Ask AI, a mode toggle reveals Buddy.

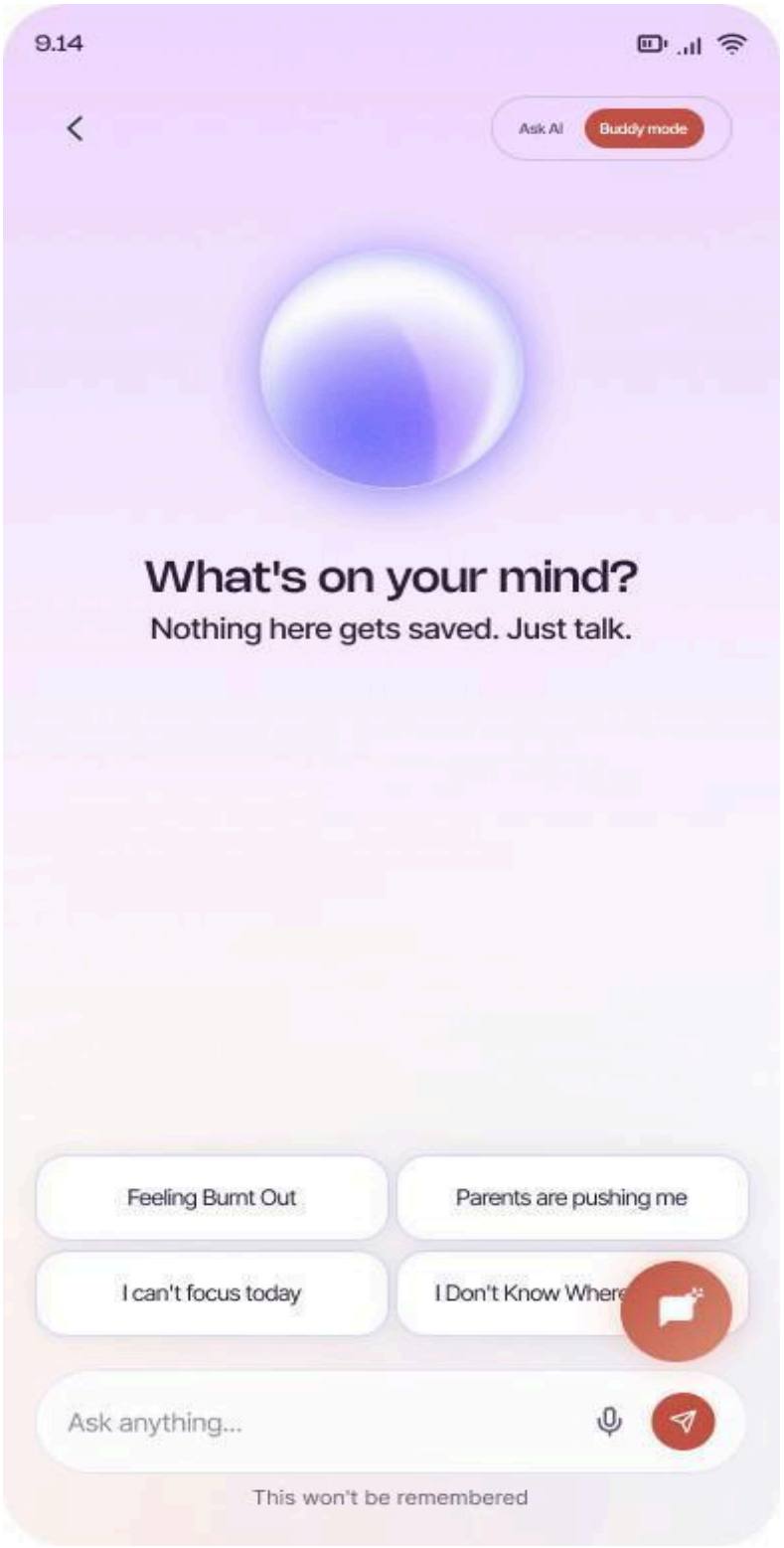


Ask AI — older-sibling tutoring voice. Hi Arjun, ask me anything.

Buddy is the peer-voice chat. Same-age tone, no therapy framing, no diagnostic prompts. The conversation can be on-record (saved to history) or incognito (saves nothing, displays a clear privacy state).



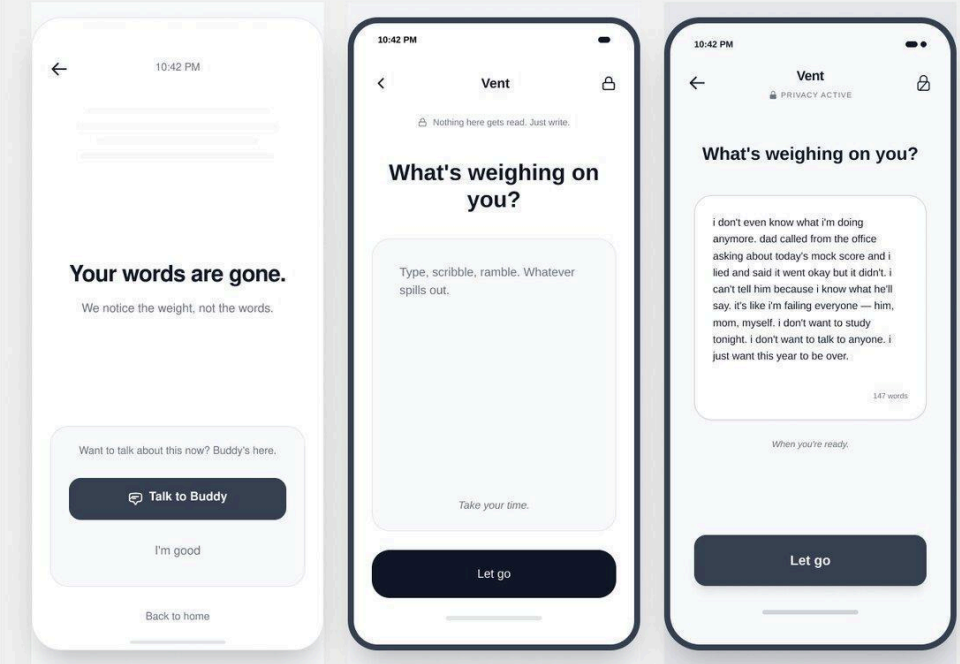
Buddy — peer-voice chat, Off The Record mode



Buddy mode — "Nothing here gets saved. Just talk." Peer-voice, no therapy framing.

Vent is a write-and-release surface. The student writes whatever is on their mind. Nothing here gets read. Nothing here gets saved unless they choose to. When they tap Let go, the words disappear. There is no AI response. There is no analysis. The release itself is the feature.

Section 9

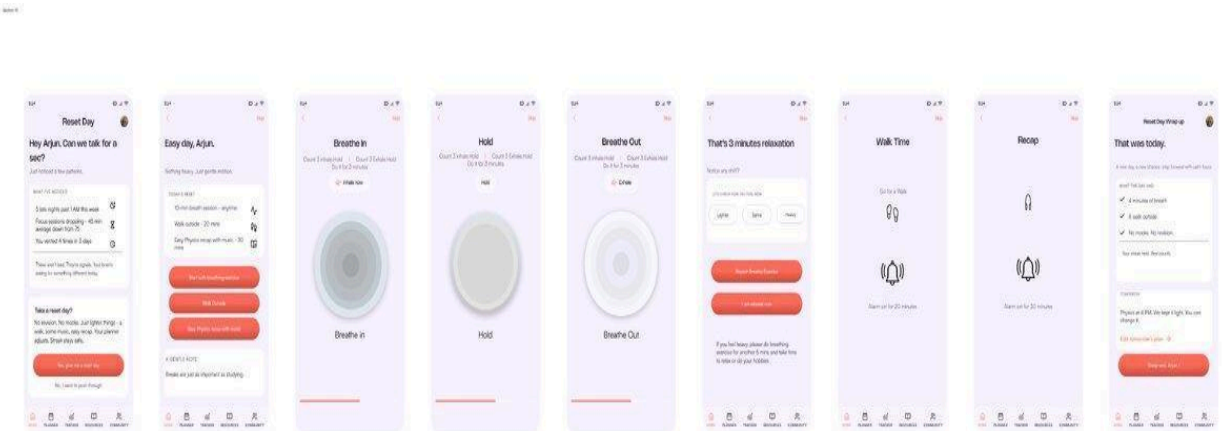


Vent — "What's weighing on you?" — write, release, gone

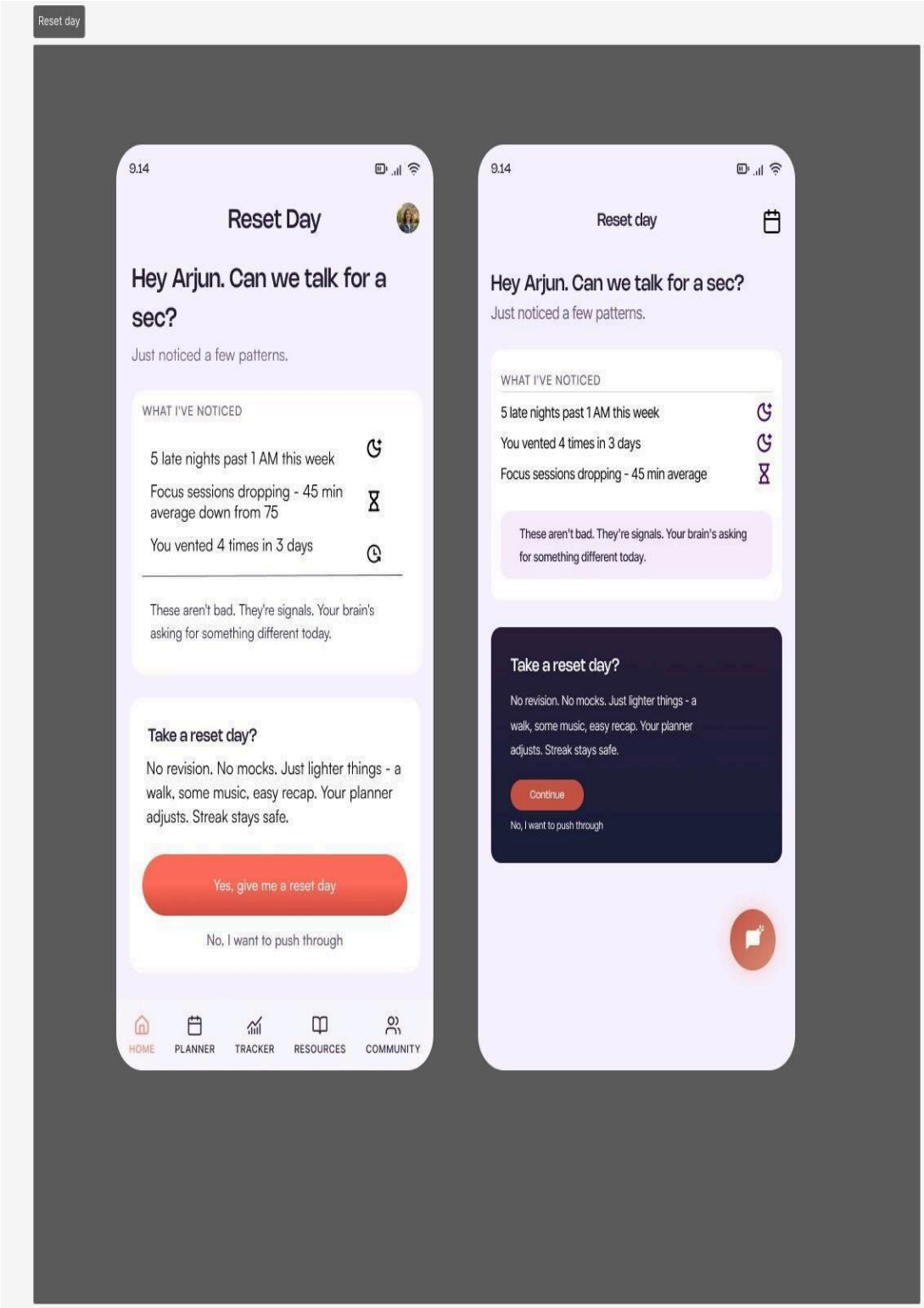


Vent Space — write what's on your mind. Nothing is saved. Nothing replies. The release is the feature.

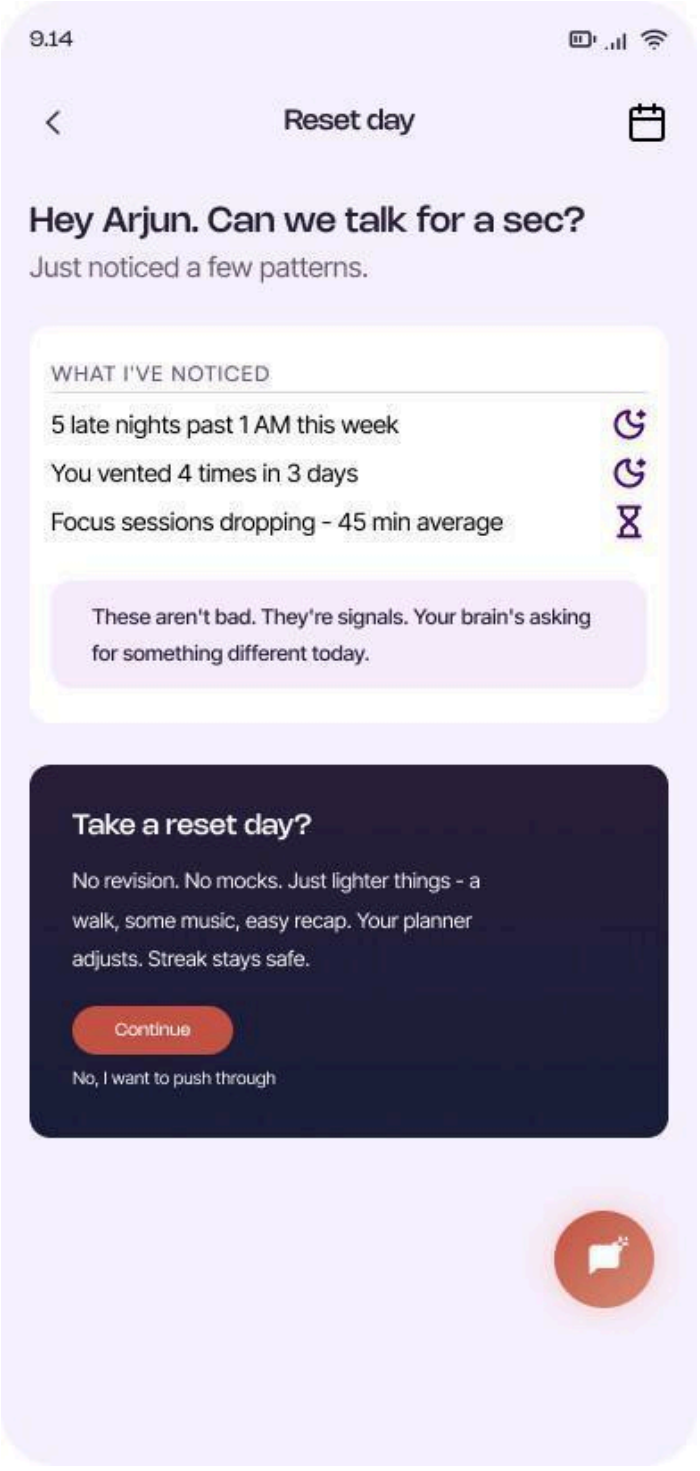
Burnout Reset triggers when the system detects missed sessions, late-night activity, or other patterns associated with the start of a spiral. Instead of guilt-trip notifications, it offers a softer alternative.



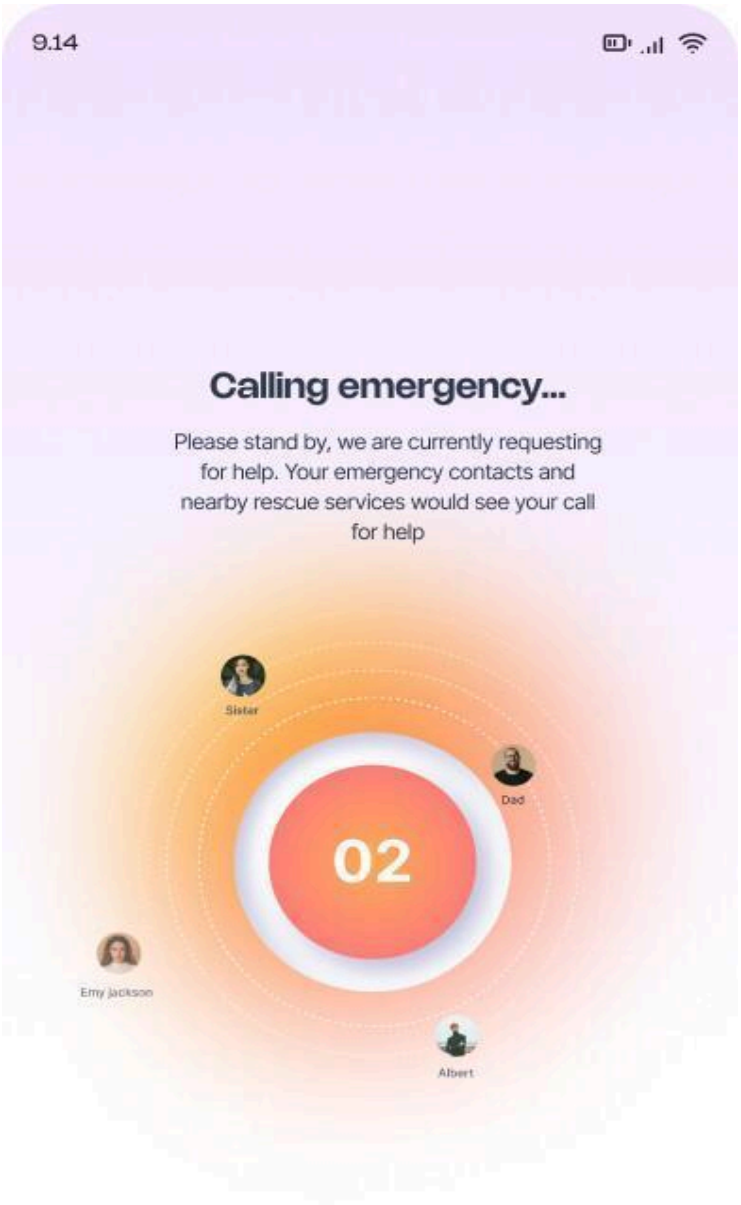
Burnout Reset — nudge card, reset day plan, breathing exercise, wrap-up



Burnout Reset — before and after screen states



Burnout Reset — Hey Arjun, can we talk? Pattern-detected nudge, soft redirect.

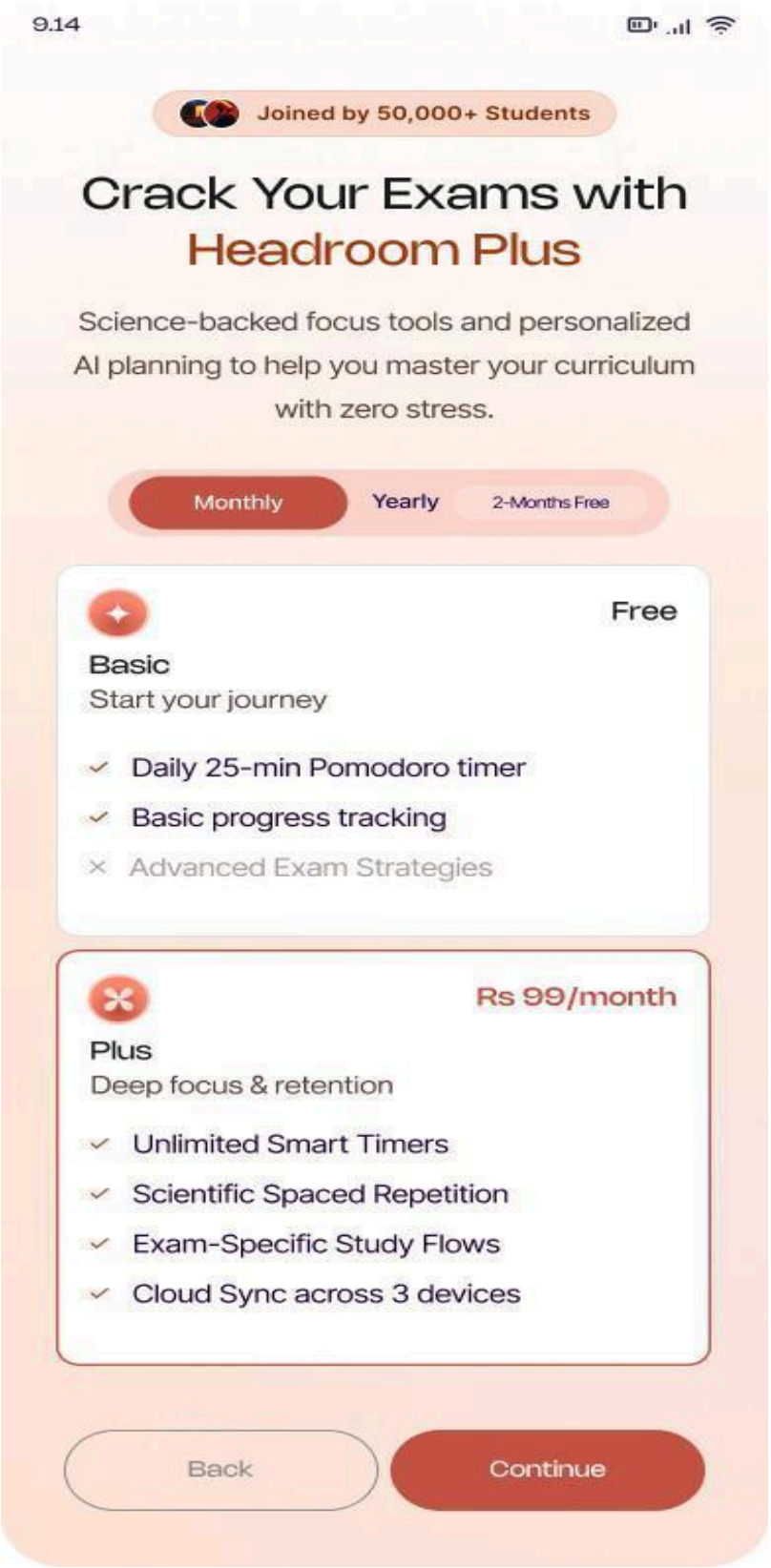


Emergency — always free, always reachable. Calling emergency contacts instantly.

The safety layer — always free, always reachable

Emergency support is accessible from the Home screen, never more than two taps from any surface. It connects the student to crisis helplines and immediate resources. Emergency support is never paywalled, never advertised on marketing surfaces, and never used as a hook to upsell. It's a baseline commitment.

Tier structure



Headroom Plus — Monthly/Yearly tier. ₹99/month, unlimited focus tools.



My Focus Status — subscription management, usage tracking, streak display.

Free tier: Ask AI (20 messages/day), planner with 1 re-plan/week, current-week tracker view, 2 Burnout Reset breaks, unlimited Vent Space, community read access + 3 posts/day, and Emergency support — always.

Plus tier (₹99/month or ₹799/year): unlimited Ask AI, smart re-planning, full tracker history, all reset breaks, advanced AI features, personal library, priority community access, premium themes.

Care tier (₹299–499/session, pay-per-session): access to verified mental health professionals, 4-session bundle option at ₹999 valid for 90 days.

SECTION 10

Reflection & Future Scope

Eight weeks. Twenty-seven research touchpoints. One hundred and twenty-two design tokens. Seven AI prompt iterations. A team of ten distributed across India. This is what we learned.

What worked

The research-first sequence held. Every product decision that survived to launch was traceable back to a specific research finding. The Trojan Horse positioning, the same-age Buddy voice, the always-free Emergency surface, the absence of mood scales on Home — none of these came from intuition. They came from interviews. The discipline of going wide in research and then narrowing in design protected us from the most common failure mode in mental-health products: solving the designer's idea of the problem instead of the user's.

The AI prototype track was the right investment. The HTML build with Groq + Llama 3.3 70B turned the most uncertain part of the product — would the AI voice actually work — into a thing we could test, iterate, and confirm. Without it, both Buddy and Ask AI would have shipped to review as speculation.

The design system held under pressure. Token Studio paid for itself within the first three days of hi-fi work. When the team needed to make global changes — a colour adjustment, a type-scale revision, a spacing token — those changes flowed across every screen automatically.

What we would do differently

- Trust team output sooner. As Captain and Lead UX, the instinct to personally review every artifact at the bar we wanted slowed the team down in the middle weeks. Future projects, that lesson is the one we carry forward most directly.
- Shorten the testing task set. The 5–6 participant drop-off rate in the Maze study was a structural signal about test length, not about the product. Two shorter sessions would have produced more complete data.
- Start the design system one week earlier. The mid-fi phase was held up briefly while the token system was being finalised. A small overlap between Visual Design and Mid-Fi tracks would have closed that gap.

Future scope

Care tier maturation. The pay-per-session counselling tier is currently structured. The next layer is partnering with verified mental health professionals at scale, building a session-booking flow, and handling the post-session continuity that students consistently asked for.

Multi-language. The current product is English-first. Research participants who were stronger in Hindi or regional languages consistently asked for native-language support, especially in the Buddy and Vent surfaces where emotional vocabulary is most personal.

Parent companion app. Parents surfaced repeatedly in research as both a pressure source and a potential ally. A parallel parent-side surface — designed against the same Glass Sanctuary principles — would extend the support ecosystem without compromising the student's privacy boundary.

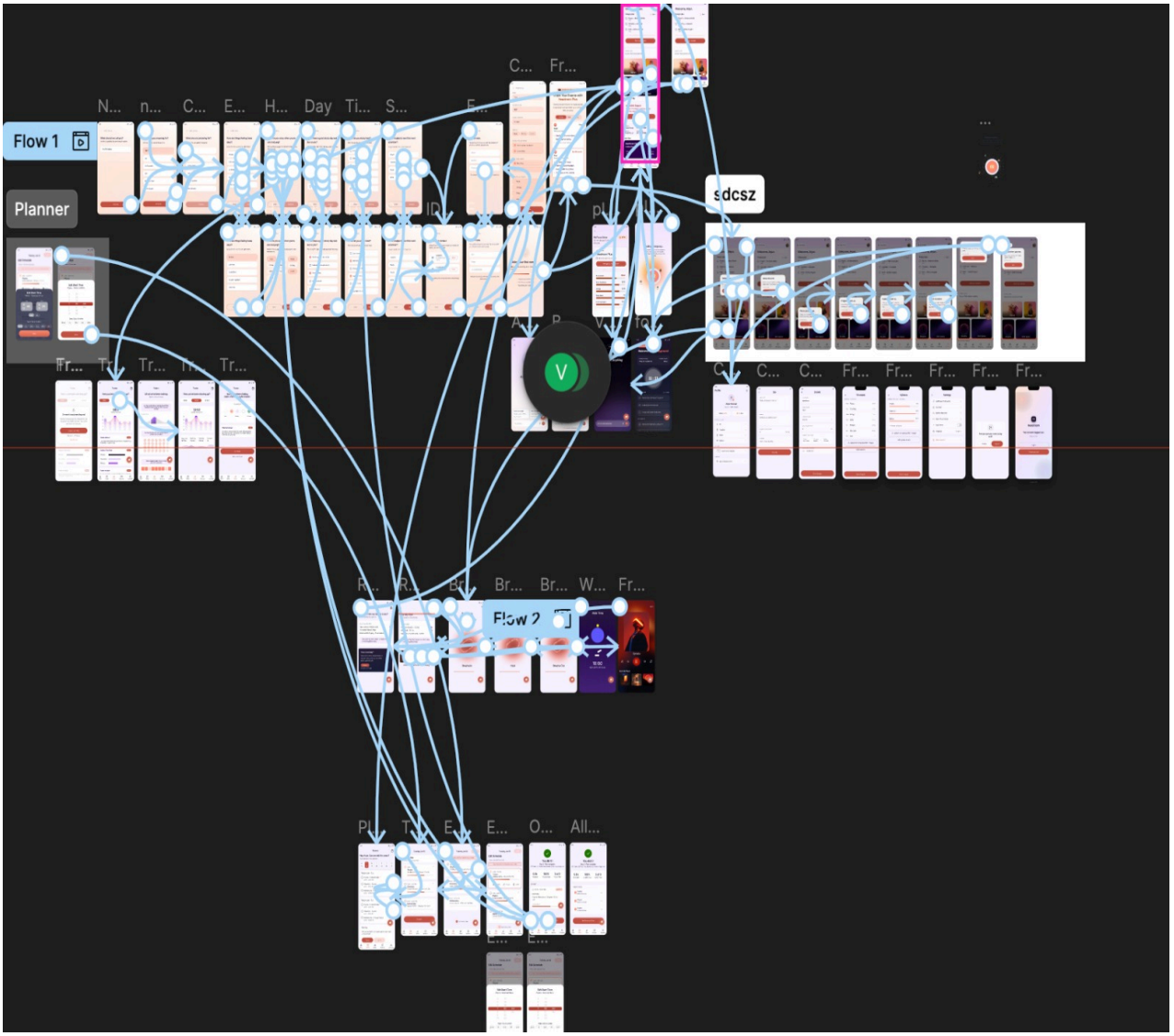
Institutional partnerships. The Government-Backed Access Model from ideation is the long-horizon play. Coaching institutes and state education boards represent the largest possible distribution path. That path opens only after the product has proven retention and outcomes independently.

> Headroom started as a question about a problem no one was solving for a population no one was reaching. The product is one answer. The work continues.

Signal Syndicate · Team 04

Hackathon 8.0 — Tesseract Frontier · June 2026

Research · Design System · Prototype · Film — built end-to-end in eight weeks by a distributed team of ten across India.



Thank you to every student who let us into a difficult conversation.